Rapid Attachment Belt-Tool (RABeT) System

Challenge 2: Lunar Surface EVA Operations – Space Suit Attachment Quick Release System

The Mechanical Engineering Competition (MEC) Team at Berkeley

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I. Technical Section

1.1 Abstract

In future lunar explorations, astronauts will be required to carry their tools on them, and will need a way to quickly and frequently access the tools. These brave men and women will face many dangers and obstacles on their missions, so it is imperative that they are set up for success in every way possible by all those involved in the preparation of these endeavors. No detail is too small, including the seemingly simple task of putting away a tool while it is not in use.

On such an inhospitable body as the moon, energy is a precious resource, therefore energy expended by the astronauts must be considered. To further complicate things, lunar soil is an incredible nuisance to everyday operations, and quite abrasive, resulting in rapid wear and potential system failures. Consequently, the Mechanical Engineering Competition (MEC) Team at Berkeley has designed a Rapid Attachment Belt-Tool (RABeT) System that can safely support tools with a weight of 45 pounds under Earth's gravity and employs continuous passive removal of lunar regolith. Furthermore, the RABeT System is designed for simple, one-handed, blind operation by the astronaut and requires minimal effort to engage and disengage, ensuring repetitive tool access does not result in unnecessary energy expenditure. The system consists of two parts: a toolside and beltside piece. The beltside piece has a large landing pad and guides to assist in aligning and mating the two pieces in one motion. Thoughtfully placed channels are designed into the RABeT System to give regolith a means to escape and avoid getting wedged between the two pieces, which would result in extra energy required to attach and separate the tool, and premature wear of the system. These details, as well as many more, are outlined in the following pages.

1.2 Design Description

The design consists of a belt-mounted interface piece and a tool-mounted interface piece that fit together with a slightly modified dovetail design which tapers from 0.565" to 0.4" in width. A stainless spring steel mounted in the beltside piece is used to prevent the pieces from separating during normal activities. Taking into consideration the material choice of aluminum, the minimum safety factor for a 45 lb load is 6.53. Figures 1.2.1 and 1.2.2 show CAD models of the toolside and beltside pieces.

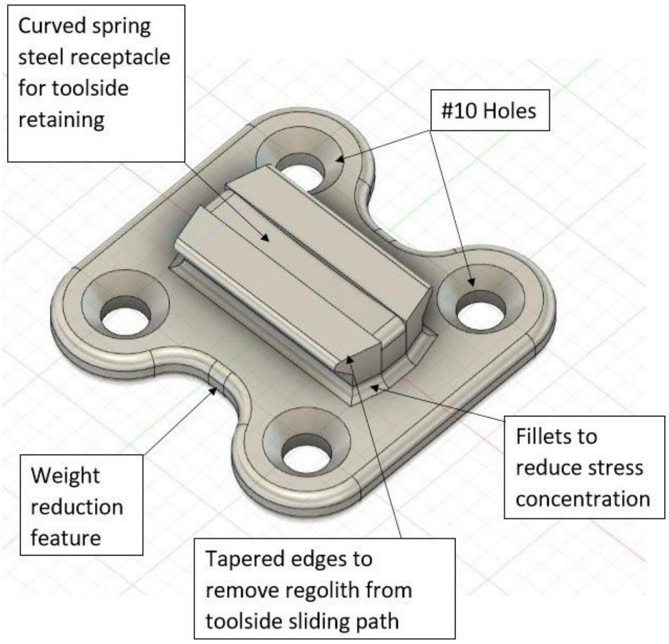

Figure 1.2.1 Toolside

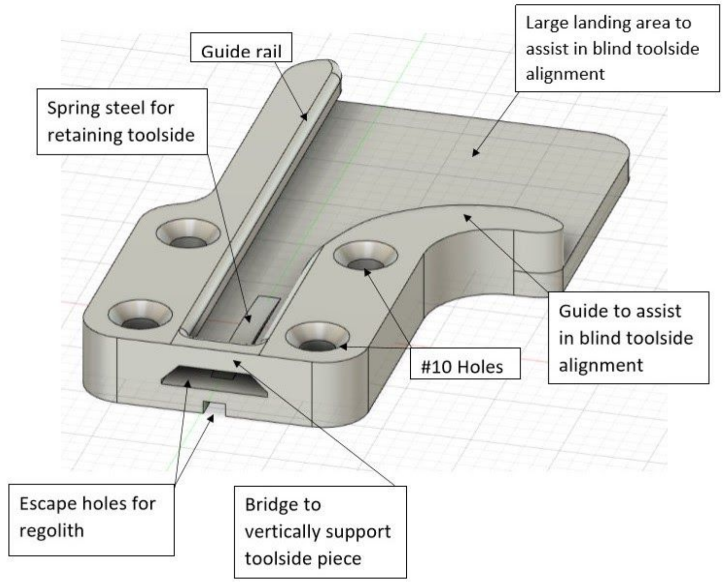

Figure 1.2.2 Beltside

1.2.1 Mating Process and Regolith Egress

The toolside piece is guided into the beltside piece by sequentially eliminating degrees of freedom. The Z direction is eliminated first by placing the toolside anywhere on the large flat surface of the beltside. Next, freedom in the X direction is eliminated by sliding the toolside until it comes in contact with the long rail that runs along the entire component. Then, the toolpiece is slid along the Y axis until it comes in contact with the bridge and is held in place by the spring steel. As an added measure, the beltside rail on the right is curved inward to funnel the toolside piece into the slot in the event it is not placed perfectly against the left rail. Figures 1.2.3 and 1.2.4 show the process for connecting the toolside piece and the beltside piece.

Figure 1.2.3 The toolside piece is aligned as it is slid into place

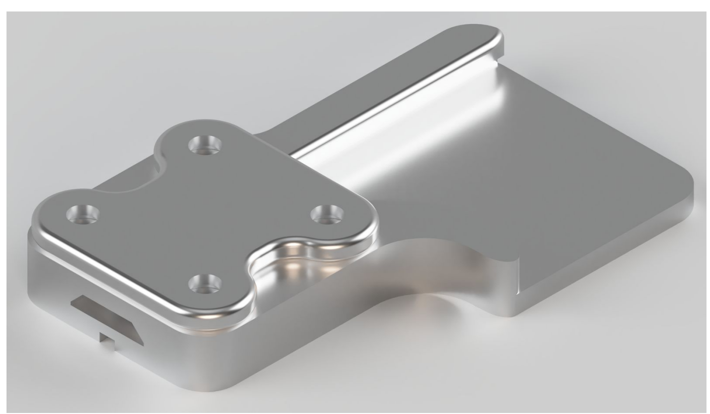

Figure 1.2.4 The toolside piece is secured

The toolside piece will be held in place by a piece of stainless spring steel. The spring steel will flex to allow the toolside piece to slide over it into place, and then it will rebound to fill the channel in the toolside piece. This channel is cut to match the spring steel when in its natural shape, so that when the tool is secured, it will not slide upwards without approximately 3 lbs of force in the positive Y direction. Figure 1.2.5 shows a section view of how the toolside interacts with the curved spring steel piece.

Figure 1.2.5 also shows two openings which have been built into the RABeT System to allow for passive removal of the regolith. The feature under the springsteel has been included to allow any regolith that gets underneath the spring steel to fall out. This will prevent the accumulation of regolith, which could compromise the functionality of the retaining device. Vertical attachment to the Utility Belt will allow regolith to be pulled out of the system by gravity through the holes shown in Figure 1.2.5 when not engaged and pushed out of the system when the two interface pieces are being mated.

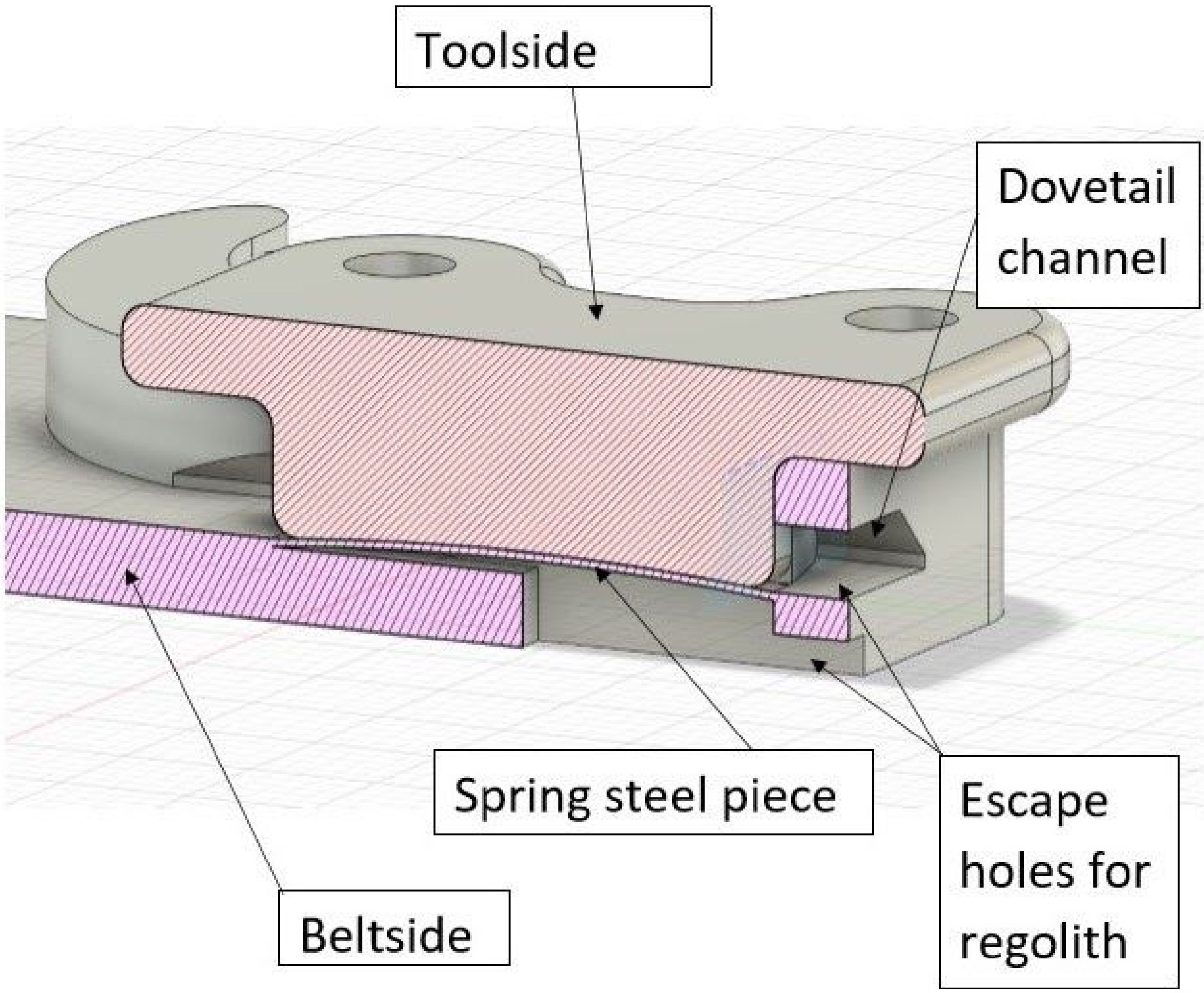

Figure 1.2.5 Spring steel interaction with toolside piece

Figure 1.2.6 shows two regolith egress routes from the RABeT System: clearance within the beltside channel and an egress hole on the bottom of the beltside piece. The 0.026” wide clearance on each side of the beltside channel created by the tapered edges of the toolpiece directs regolith out of the toolside’s path when mating the tool to the Utility Belt, giving space for the regolith to exit the RABeT System through the openings at the bottom.

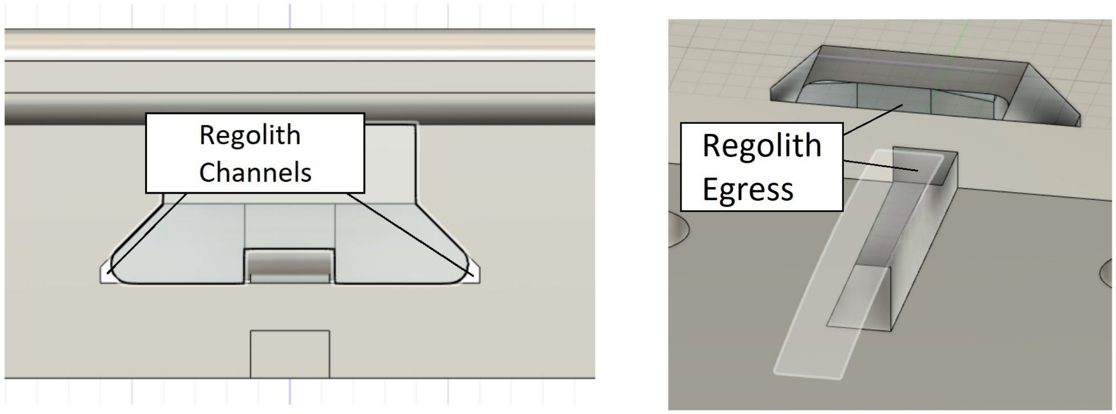

Figure 1.2.6 Channel Clearance and Regolith Egress Holes

1.2.2 Manufacturing Plan

The toolside will be milled down from a single piece of aluminum using a 5-axis CNC mill. The beltside only requires a 3-axis CNC mill, and it will be manufactured in two components. These components will be joined together via the #10 screws which also attach the piece to the Utility Belt. Figure 1.2.7 shows the two components which will comprise the beltside.

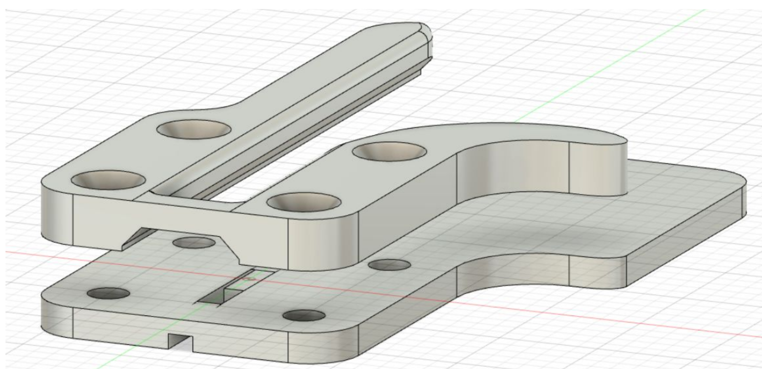

Figure 1.2.7 Beltside piece

Typically, the team would be responsible for manufacturing the RABeT System using the Makerspace and the Machine Shop made available to students by the University of California, Berkeley. Some members of the team have basic machine shop training with lathe and mill, and those members of the team would be responsible for manufacturing the system. However, due to COVID-19, student use of the spaces has been limited and it may be necessary to give the design to the machine shop staff at UC Berkeley to evaluate and manufacture. The design will be further revised and finalized with the supervision of the NASA Micro-g NExT mentor and the faculty advisor before final manufacturing drawings are produced. The team does not anticipate needing to train members who do not have sufficient manufacturing skill to manufacture the system. However, if it is needed, those team members will acquire training from the machine shop staff. The team's manufacturing schedule with assignments is presented in Figure 1.2.8. Each team member will undertake tasks they are capable of accomplishing.

TASK NAME	START DATE	END DATE	START ON DAY*	DURATION* (WORK DAYS)	TEAM MEMBER	PERCENT COMPLETE
Design finalization						
Conversation with NASA mentor and Faculty	1/13	2/1	0	19	Pranit, Trey	0%
Revise and finalize design	1/20	2/5	7	16	Anthony, Pranit, Todd	0%
Finalize manufacturing plan	1/20	2/12	7	23	Wendy	0%
Create manufacturing drawings	1/27	2/12	14	16	Anthony	0%
Learn manufacturing skills	2/1	2/5	19	4	Anthony, Wendy	0%
RABeT System Manufacturing						
Contact machine shop and makerspace	1/21	1/30	8	9	Anthony	0%
Contact with ASUC finance	1/21	2/1	8	11	Trey	0%
Tool manufacturing	2/15	3/22	33	35	Anthony	0%
Tool evaluation	4/1	4/9	78	8	Pranit	0%
Preparation and Testing						
Plan Agenda for Meeting	1/20	5/6	7	106	Trey	0%
Prepare for testing	4/12	4/26	89	14	All team	0%
Finalize Tool Presentation	4/27	5/7	104	10	Wendy	0%
Testing at NBL	6/7	6/12	145	5	All team	0%
Documenting Lesson Learned	6/14	6/18	152	4	All team	0%

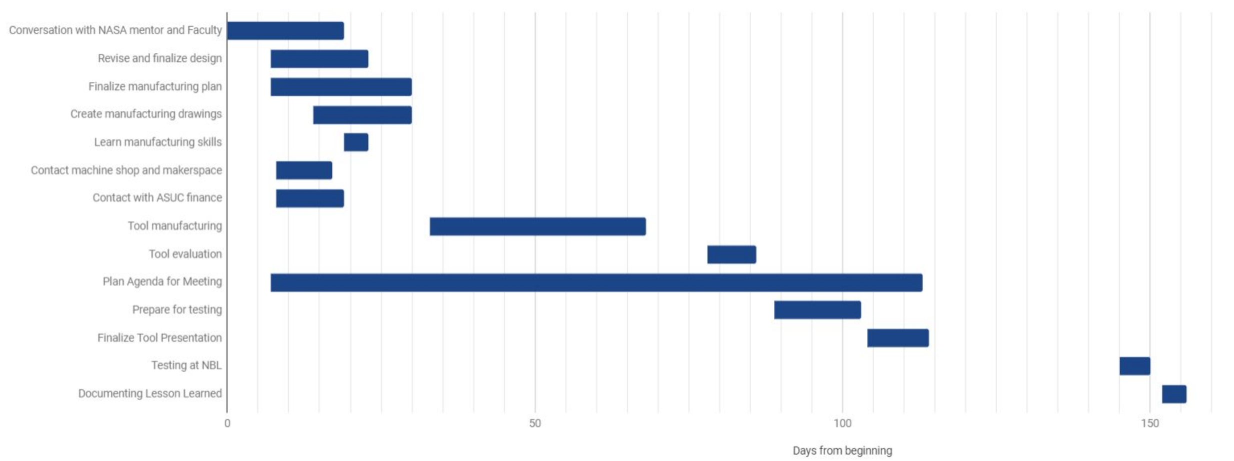

Figure 1.2.8 Team manufacturing schedule and Gantt chart

Figure 1.2.9 shows two-dimensional sketches of the RABeT System to be used for manufacturing.[1] It also shows cutaways revealing details of the spring steel and regolith egress routes.

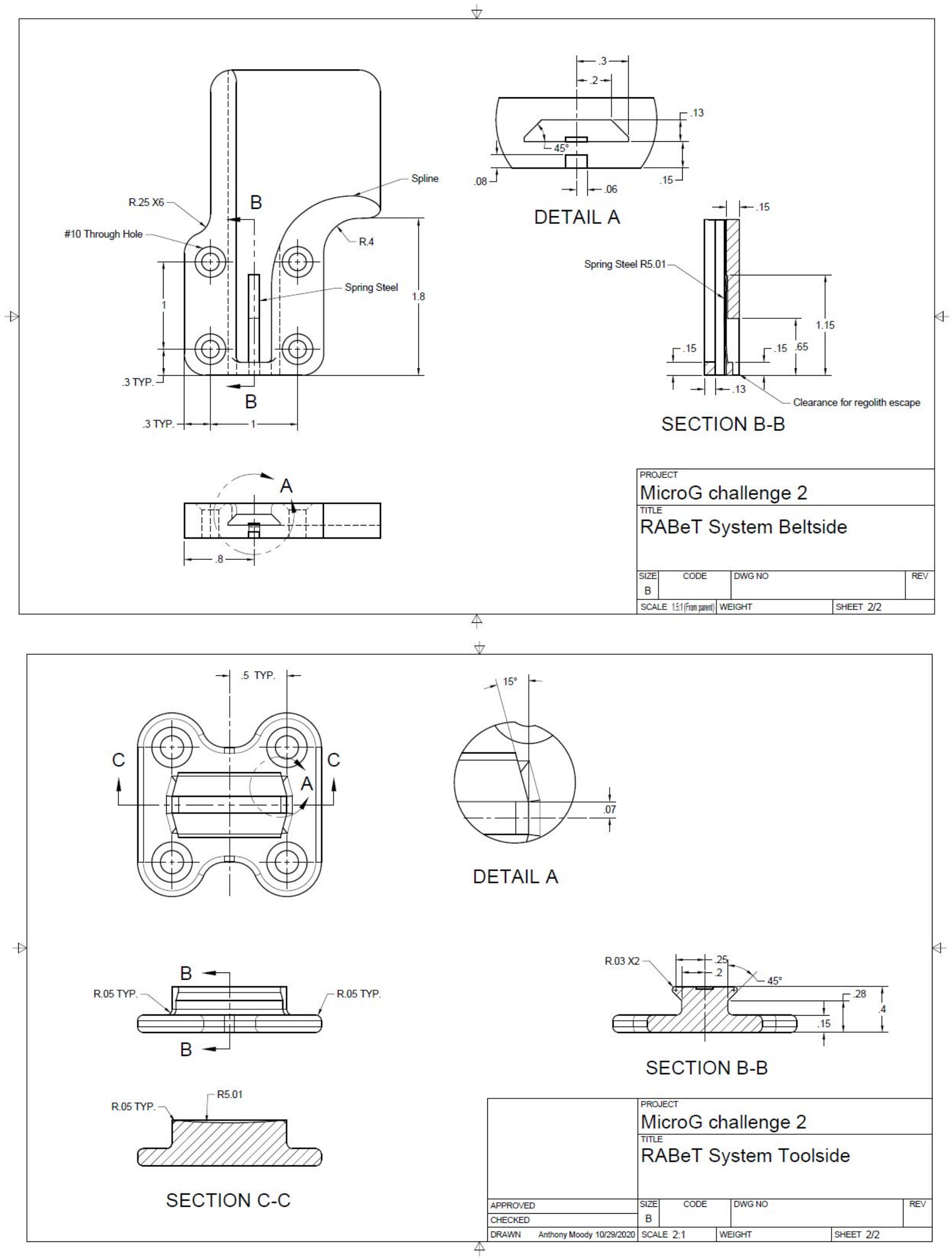

Figure 1.2.9 2-Dimensional Drawings

[1] Additional technical drawings found in Appendix 4.1

1.2.3 RABeT System Analysis

The current state-of-the-art design used on the ISS EVA uses a thin bayonet fitting on tools required for EMU (Trevino). This is not desirable for lunar operations as dust can accumulate easily on the bayonet receptacle. During Apollo missions, astronauts used tool carriers to carry their tools—except for their camera, which was attached to the EMU suit via a large bayonet fitting ("Apollo," United States 39-46). The fitting incorporated an upside down receptacle to prevent dust accumulation. However, it is large and cannot be fitted on small tools (Thomas). In contrast, the design presented here is applicable to tools ranging in size from 1.5 by 1.5 by x inches up to any reasonably sized tool, and they can weigh as much as 45 pounds (under Earth's gravity) while maintaining a minimum safety factor of 6.53. The final design compliance with the requirements is listed in the compliance matrix (Figure 1.2.10):

Requirement Compliance Matrix			
Requirement Num.	Description	Compliant	Comments
1	The attachment shall be able to support at least 15 lbs of weight in Earth's gravity.	Yes	45 pounds with Safety Factor of 6.53
2	The attachment mechanism shall be operable outside of the astronaut's line of sight. In other words, the astronaut shall be able to operate the mechanism without being able to see it.	Yes	Multi-axis guides for blind alignment
3	The attachment mechanism shall be operable with one hand (the other hand can be used to place the tool into the mechanism).	Yes	Operable with one hand on the tool only
4	The device shall be able to function as intended after being fully submerged in lunar dust simulant. The device will be fully submerged in lunar dust simulant, removed, and then cycled 10 times	Yes	Passive removal of regolith, minimal moving parts
5	The two pieces should install and separate from each other with minimal force for ease of use but has enough force to stay in place while walking/bending	Yes	Gravity, and stainless spring steel retainer
6	The device shall use only manual power	Yes	
7	The device shall fit within a volume of 4" by 4" by 3", the smaller the better	Yes	Total Dimensions 3.5" x 2.25" x 0.55"
8	The device shall have a 4-hole bolt pattern to interface with the Utility Belt. See Interface Details below	Yes	
9	The device must be operable with EVA gloved hands (like heavy ski gloves)	Yes	
10	The total weight of all parts you provide should be less than 2 lbs, the lighter the better.	Yes	Total Weight 0.22 lbs (Earth gravity)
11	The parts must not have holes or openings which would allow/cause entrapment of fingers	Yes	
12	The parts should be made from only Aluminum 6061, Aluminum 7075, Stainless Steel (any series), or Teflon. Any other materials must get prior approval from the EVA Tools POC*.	Yes	Stainless Spring Steel, Aluminum 6061
13	There shall be no sharp edges on the tool.	Yes	All edges filleted
14	Pinch points should be minimized and labeled.	Yes	1 pinch point

Figure 1.2.10 RABeT System compliance matrix

1.2.4 Design Process

Initially, the team settled on three potential high-level concepts. The first was a belt clip system (similar to that found on a tape measure), into which a rod attached to the tool would slide in. The second was a self-mating c-clip and magnet combination inspired by Fidlock buckles that utilized a magnet to assist the user in aligning and mating the beltside and toolside pieces. The third option, which would become the first version of the final concept, consisted of a toolside slider with four possible orientations, that would slot into a complementary beltside rail, and click into place due to spring steel.[2]

To aid in the decision making process, a Pugh Chart was made.[3] Criteria specifically provided by the challenge requirements were weighted most heavily; items such as part cost and manufacturability were weighted least heavily since these costs are nearly negligible relative to launch costs. Strength encompassed not just nominal load cases, but also the case of an off-nominal load case (an accidental lateral impact to the belt-side for example). All potential concepts were expected to perform well in terms of ease of actuation, safety, and tool security. However, since lunar soil is more than 10% Iron (II) Oxide in some areas (Heiken 365), the team anticipated major issues with dust accumulation for the c-clip plus magnet design. It was decided that even if the overall scores indicated this design would be the best option, the susceptibility to failure from regolith accumulation virtually disqualified the design. The belt clip design grossly failed to meet the stability requirement issued in the challenge and was similarly disqualified. The RABeT System was clearly the best design of the three, and the Pugh chart supported that conclusion, so the team chose to proceed with that design.

After the general concept was determined, the design went through further iterations. Initially, the toolside was designed such that it could be mated to the beltside piece in 4 different orientations (each separated by 90 degrees); for tool stability and ease of actuation this was compromised down to two orientations, separated by 180 degrees. The initial beltside design was symmetric, with both rails angling away from the centerline to create a funnel for the toolside piece; this was changed to an asymmetric design to provide an extremely easy alignment method so the astronauts could conveniently slot in the tool, even without being able to visually verify alignment. Other changes were made to enhance the viability of the design that are outlined in the design description.

1.3 Operations Plan

The testing operations plan at the NBL shall take approximately 30 minutes, with 18 minutes allotted for procedures and another 12 minutes allotted for transitioning between procedures and in-test feedback from the tester. The operations plan is presented as follows:

1. Start with Utility Belt and tool interface pieces mated together
   - 1.1. The tool interface piece shall be inserted into the Utility Belt interface piece as described in the design description and pushed all the way down until it makes contact with the stop.
   - 1.2. This will be the starting point for each test.
2. Get a feel for the system (2 minutes total)
   - 2.1. Using one hand only on the tool, remove the tool and reattach it ten times
   - 2.2. Evaluate if it is difficult to cycle using only one hand on the tool
   - 2.3. If so, one hand on the tool and one hand on the Utility Belt may be used throughout the remainder of the testing session

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[2] Sketches of each design found in Appendix 4.2

[3] Found in Appendix 4.3

2.4. If two hands are necessary, cycle through the attachment process ten more times using one hand on the tool and one hand on the Utility Belt around the Utility Belt interface piece as needed
2.5. Attach the tool to the belt and proceed to the next test

3. Stability and security (5 minutes total)
   3.1. General stability (1 minute)
      3.1.1. Place hand on the tool and move it around to evaluate the amount of play in the system
      3.1.2. If there is any play, evaluate how much the parts of the tool that are farthest away from the center of the system move
      3.1.3. Confirm tool is properly attached and proceed to part 3.2
   3.2. Tool Security when bending down (2 minutes)
      3.2.1. Bend down as if to pick something up off the ground and stand back up
      3.2.2. Evaluate if the tool remains attached and in its original position
      3.2.3. Repeat ten times
      3.2.4. Confirm tool is properly attached and proceed to part 3.3
   3.3. Security when walking (1 minute)
      3.3.1. Walk around for thirty seconds
      3.3.2. Evaluate if the tool remains attached and in its original position
      3.3.3. Confirm tool is properly attached and proceed to part 3.4
   3.4. Security when hopping (1 minute)
      3.4.1. Jump up and down ten times
      3.4.2. Evaluate if the tool remains attached and in its original position
      3.4.3. Confirm the tool is attached and proceed to the next test
4. Evaluate how well the tool functions covered in regolith simulant (This may not be a good test for the NBL since the water will likely wash away the simulant - A similar test will be performed out of the water) (3 minutes total)
   4.1. Remove the tool
   4.2. Drop the tool in regolith simulant, making sure to cover the tool attachment piece in simulant
   4.3. Recover the tool and cycle through the attachment process ten times
   4.4. Evaluate if the process feels any different than in procedure 2 - If yes, report how so
   4.5. If so, repeat the steps 3.1 to 3.4, but give the tool a couple shakes to try to remove any simulant before cycling
   4.6. Evaluate if step 4.5 improves the process
   4.7. Attach the tool to the Utility Belt and proceed to the next test
5. Attempt to attach the tool incorrectly (2 minutes total)
   5.1. By wedging the long side of the tool interface piece down into the Utility Belt interface piece (1 minute)
      5.1.1. Remove the tool and rotate it 90 degrees so that the vertical plane of the tool interface piece remains parallel to the vertical plane of the Utility Belt interface piece
      5.1.2. Attempt to mate the pieces with the same motion that would be used if the tool was in the correct orientation
      5.1.3. Repeat five times
      5.1.4. Evaluate if it is possible to get it stuck
      5.1.5. If so, evaluate how difficult is it to dislodge the pieces
      5.1.6. Attach the tool to the belt and proceed to part 5.2
   5.2. By sliding only one rail into the belt piece slot (1 minute)

5.2.1. Remove the tool and rotate it 45 degrees clockwise so that the vertical plane of the tool interface piece is at a 45 degree angle to the vertical plane of the Utility Belt interface piece
5.2.2. Try to mate the pieces using the correct motion, but at the new angle
5.2.3. Repeat five times
5.2.4. Evaluate if it is possible to improperly attach the pieces in such a way
5.2.5. If so, evaluate if they come apart easily
5.2.6. Attach the tool to the belt and proceed to part 5.2.7
5.2.7. Remove the tool and rotate it 45 degrees counterclockwise so that the vertical plane of the tool interface piece is at a 45 degree angle to the vertical plane of the Utility Belt interface piece
5.2.8. Try to mate the pieces using the correct motion, but at the new angle
5.2.9. Repeat five times
5.2.10. Evaluate if it is possible to improperly attach the pieces in such a way
5.2.11. If so, evaluate if they come apart easily
5.2.12. Attach the tool to the belt and proceed to the next test

6. Evaluate mating pieces without visual reference (2 minutes total)
   6.1. Remove the tool
   6.2. Bring the tool down to the side and relax arm
   6.3. Pause for five seconds
   6.4. Attach the tool to the Utility Belt without looking at it
   6.5. Repeat 6.1 through 6.4 ten times
   6.6. Evaluate how difficult is it to mate the pieces without looking at them
   6.7. Proceed to the next test
7. Multi-directional attachment (2 minutes total)
   7.1. Remove the tool
   7.2. Invert the tool so that its position is rotated 180 degrees in the vertical plane
   7.3. Attach the tool in this orientation
   7.4. Repeat 7.1 through 7.3 ten times
   7.5. Evaluate if the system works the same as it did in the original orientation
   7.6. Remove the tool and reattach it in the original orientation then proceed to the next test
8. Strength (2 minutes total)
   8.1. Use the tool as a lever to try to deform the quick release system (with a reasonable amount of force to simulate getting caught on something) *Do this last in case of deformation
   8.2. In the event of deformation,
      8.2.1. Evaluate the state of deformation
      8.2.2. Attempt to cycle through the attachment process ten times to determine if the system still functions with deformations
      8.2.3. Record any difference in performance
   8.3. Attach the tool.
   8.4. Return to surface.

1.4 Safety

A myriad of precautions have been taken in designing the RABeT System to limit possible hazards. They are summarized below.

1. All edges will be filleted to ensure there are no sharp edges on the RABeT System.
2. A pinch point exists where the tool interface piece is inserted into the Utility Belt interface piece. However, the system has been designed in such a way that the astronaut only needs to have one hand on the tool to attach it to the Utility Belt. The other hand will remain free and clear of the device. As a result, the risk of pinching a glove is minimal. In the event that the astronaut needs two hands to operate the system, having one hand on the tool and one hand stabilizing the system, the astronaut shall hold the Utility Belt around the Utility Belt interface piece instead of the piece itself to make certain the glove does not get caught in the pinch point. Astronaut training on the proper use of the system will be performed to further mitigate the likelihood of pinching.
3. Care has been taken to ensure there are no holes or openings in the device that would allow or cause entrapment of fingers.
4. Risk of lunar dust accumulation is eliminated via the choice of nonmagnetic materials to manufacture the system as well as incorporating a passive removal system for the regolith in the design.
5. Risk of system failure was mitigated through Finite Element Analysis (FEA) and redesigns based on the results.
    - 5.1 A later version of the RABeT system featured an extrusion at the bottom of the beltside piece to stop and support the toolside piece when mated. This was the weak point in the design, but the design still had a safety factor of 11.26 for a 15 lb (Earth's gravity) static load. In later iterations when the channel beneath the spring steel was added, a major weak point was created and the safety factor dropped to less than 2. In response, the extrusion was replaced with a bridge connecting the left and right halves of the tool-facing portion of the beltside piece resulting in a safety factor of 15. The upper limit of the safety factor in the FEA software (Autodesk Fusion 360) is set at 15, so the actual safety factor could very well be higher than that. See Appendix 4.4.1 for final FEA results with 15 lb load.
    - 5.2 Further FEA analysis determined the system can support a static load of 45 pounds of weight in Earth's gravity with a minimum safety factor of 6.53. See Appendix 4.4.2 for final FEA results with 45 lb load.

An FMEA (Figure 1.4.1) has been created to evaluate the risks of the RABeT System and present possible mitigation techniques.[4]

Item	Failure Mode	Potential Causes	Effects of Failure	Mission Impact (A-E)	Likelihood (1-5)	Risk Categorization	Recommended Action	Responsible Parties	Actions Taken
1	Tool Detachment	Insufficient attachment between tool and buckle	Tool falls out of buckle	A	2	B2	Strengthen locking mechanism	Design team	Added spring-steel locking mechanism
2	Suit cutting	Sharp edges	Suit leakage	E	1	E1	Remove any sharp edges	Design team	All edges are filleted on the tool
3	Buckle breaking	Too much force, too heavy	Broken buckle	D	1	D1	Limit tool weight; strengthen buckle	Tool department	Widened support for tool so safety factor is >15
4	Finger entrapment	One hole exists	Glove trapped	D	1	D1	Mitigation training	Astronaut training team	Minimized number of possbile entrapment points
5	Glove pinching	One pinch point exists	Glove pinched	D	1	D1	Mitigation training	Astronaut training team	Minimized number of possible pinched points
6	Tool misses belt connection	Insufficient area to attach tool and buckle	Tool not secured	A	3	A3	Mitigation training	Astronaut training team	Created large surface for astronaut to line up one surface before sliding tool to axis-aligning rail
7	Moon Dust Accumulation	Magnetic materials, small volume in tool	Tool stuck in buckle	C	2	C2	Eliminate small volumes in tool	Design team	No magnetic material used in tool; added holes so moon dust can exit buckle

Figure 1.4.1 FMEA of the RABeT System

1.5 Technical References

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[4] FMEA Risk Matrix criteria found in Appendix 4.5

II. Outreach

2.1 Plan

Through the outreach plan, the team's goal is to inspire the next generation of scientists and engineers. Each team member remembers a specific instance that gripped their curiosity, inspiring them to pursue a degree in engineering. The MEC Team hopes its program may serve as this spark for the students they teach.

2.1.1 Workshop Program

The team focused on a workshop program for 6-12 grade level students as the main portion of the outreach plan. This target audience was chosen because in the structured school curriculum, students are rarely exposed to real-life engineering challenges; rather, they are mainly taught the building blocks which they put together later in their education. Students in grade levels 6-12 are in the period where they are considering their future career path, and they are ready to take on a deeper level of science and engineering. The team hopes to inspire these students to pursue STEM by guiding them through several design challenges.

The team's workshop program has two main objectives:

1. To teach 6-12 students the engineering project planning process, similar to the process the team goes through in this Micro-g NExT challenge, and
2. To teach 6-12 students the engineering challenges of designing systems that would be used on the moon, similar to the engineering challenges the team has to undergo to complete the Micro-g NExT challenge

Each of the team's lesson plans addresses both objectives. The goal is that each of the lessons serves as a foundation for the students to pursue a career in aerospace, be it in STEM, finance, or politics. To further this goal, the team specifically reached out to underprivileged or low performing schools around the team members' communities (Alameda, Sacramento) where STEM opportunities are not readily available, as well as schools with which team members had previous connections. This was accomplished by researching schools on the California Comprehensive Support and Improvement list in each of the team members' respective communities and presenting them with the team's lesson plan. The schools then decided if they would welcome the team to perform the lesson plan.

In total, the team contacted nine schools asking if they would be interested in the team leading a workshop with their students. The team is still in the process of finalizing the dates with school faculties to lead the workshops, but they plan to finalize them by the end of November.

2.1.2 Publicity Plan

In addition to leading workshops with 6-12 students, the team hopes to inspire others by sharing their successes with their peers, the general public, and middle and high school students.

The team wrote a press release[5] which—if selected for Phase Two—they will send to local newspapers such as *The Daily Californian*, *Berkeleyside*, and *The East Bay Times*, spreading the word about their journey and hopefully inspiring others to embark on similar journeys.

[5] Found in Appendix 4.6

The team has contacted multiple UC Berkeley podcasts and a UC Berkeley TV program about possible interviews if they are selected for Phase Two. The team will share their story and describe their design process.

In addition to these, the team plans to chronicle their journey through the NASA Micro-g NExT Challenge on their newly created Facebook page available at https://www.facebook.com/Mechanical-Engineering-Competition-Team-at-Berkeley-112353670661360. The team hopes to inspire others by showing how they are able to overcome obstacles they face while continuing to improve their design.

2.1.3 Implementation of Objectives

The team's outreach activities are summarized in Figure 2.1.1. This table is a working document and it contains completed and planned activities. The team's outreach schedule is shown in Figure 2.1.2.

Date	Type of Outreach	Organization(s)	Audience	Status
Oct. 2020	Middle School Workshop	St. Ignatius Parish School	~120 middle school (6-8) students	Received letter of confirmation,[6] finalizing date
Oct. 2020	High School Workshop	Monterey Trail High School	~20 high school (9-12) students	Received interest,[7] currently determining dates for workshop
Oct. 2020	Middle School Workshop	Anthony W. Ochoa, Bret Harte, Elmhurst United, Frick, James Rutter, Westlake, Will C. Wood	Seven Middle Schools	Contacted[8]
Oct. 2020	Newspaper Press Release	*The Daily Californian*, *Berkeleyside*, *The East Bay Times*	General public of Berkeley and the East Bay	Press release written, will be sent if selected for Phase Two
Oct. 2020	Podcast Interview	*Fiat Vox*, *The (Not So) Secret Guide to Being a Berkeley Engineer*	UC Berkeley community	Contacted
Oct. 2020	TV Interview	*CalTV*	UC Berkeley community	Contacted
Oct. 2020	Social Media	N/A	Followers of the Facebook page	Facebook page created, will be updated along the team's journey

Figure 2.1.1 Outreach Plan Progress

[6] Found in Appendix 4.7
[7] Communication found in Appendix 4.7
[8] Email template found in Appendix 4.8

MEC Team Outreach Schedule

TASK NAME	START DATE	END DATE	START ON DAY*	DURATION* (WORK DAYS)	TEAM MEMBER	PERCENT COMPLETE
Outreach						
Workshops with middle and high schools	11/15	3/5	0	110	All Team	0%
Interviews with podcasts and TV show	12/1	1/1	16	31	Trey	0%
Send out press release	12/1	12/2	16	1	Todd	0%
Update Facebook account	11/15	6/12	0	209	Todd	0%

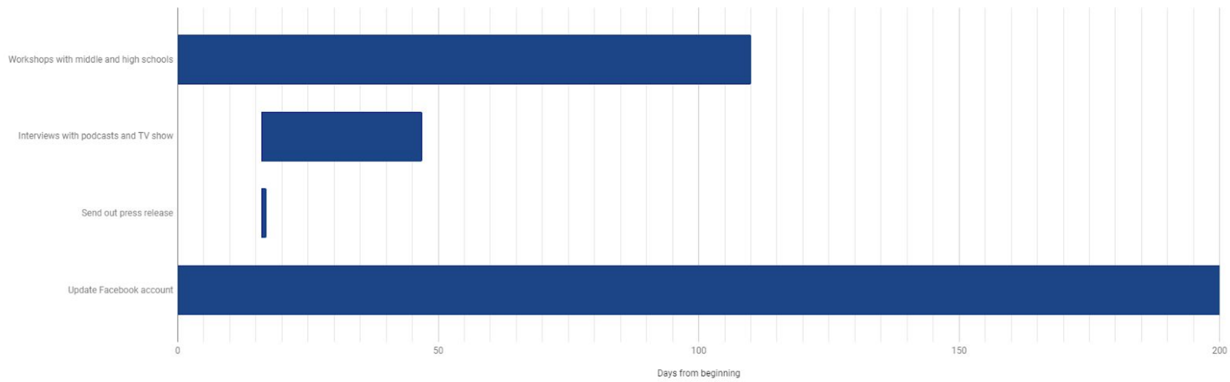


Figure 2.1.2 Team Outreach Schedule

2.2 Activities

The team's lesson plans are divided into two categories: 6-8 (middle school) and 9-12 (high school). The middle school lesson plan was created in a way to keep middle school student's attention while still teaching them pertinent engineering concepts. The high school lesson plan contains a deeper level of language which is more representative of the college and industrial engineering career paths. Each of the lesson plans cover a few California Educational Standards. Due to the current COVID-19 pandemic and California regulations, the team's outreach efforts are limited to virtual workshops; however, the lesson plans contain options for hands-on development if the current pandemic restriction is lifted or if school faculty members want to use the lesson plan for further development. The individual lesson plans are described below.

2.2.1 Middle School (6-8)

The Lunar Design Lesson Plan[9] is an hour-long workshop which teaches middle school students about the science of the environment of the moon, what engineering challenges face human lunar missions, and the current missions planned for the moon. After learning these concepts, the students will then construct an apparatus to shield a human model from the effects of lunar dust, taking into consideration what they learned in the lesson.

Upon arrival, MEC Team members engage the students by discussing their experience designing a tool for the Micro-g NExT Challenge, describing their project planning process. Team members will then lead a presentation explaining past human exploration of the moon, future human exploration, and the

[9] Full lesson plan found in Appendix 4.9

challenges of the moon. Team members will elaborate on the challenges of exploring the moon, examining challenges such as lack of air, lunar dust, reduced gravity, radiation, and cold temperatures. These items address California Educational Standards CTE.EA.B.4.0, MS-ESS1-2, and MS-ETS1-1.

Team members will then guide the students as they explore these concepts in a hands-on activity designing and constructing an apparatus to shield a human model from the effects of lunar dust, evaluating what the students have learned from the lesson. Students will be given cardboard, glue, and tape and their construction materials, and they must design within certain constraints including but not limited to: Their apparatus must have a way for astronauts to get into and out of it, it must be under a certain size, and/or it must be under a certain weight. Team members will draw upon their experience designing a tool which must operate in the presence of lunar dust to help mentor the students.

After the students construct their apparatus, team members will evaluate its effectiveness by carefully pouring sand over it, examining how much sand gets inside.

2.2.2 High School (9-12)

The Water Design Lesson Plan[10] is a week-long workshop devised to teach high school students about the design process of an engineering project through five lessons. Students will learn how to develop specifications, how to develop a tool that conforms to specifications, and how to evaluate and test the produced tool. This lesson plan addresses California Educational Standards CTE.EA.B.7.0, CTE.EA.C.2.0, and MS-ETS1-1.

In the first lesson, MEC Team members engage the students by introducing systems engineering as a key role in projects ranging from small items the students see everyday such as handheld tools to large systems the students know about and hold in awe such as the International Space Station. Team members will explore how systems engineering is vital to project success. Team members will then pose the design problem to the students: develop a low-cost tool to be operated by divers cleaning trash out of the Sacramento River. The design problem is constructed so it poses a challenge which is similar to the lunar design challenge but applied to an Earth-based problem. Team members will explain the concept of stakeholder expectations and elaborate on how the students can meet them. At the end of the lesson, team members will pose a set of questions for the students to work on as homework which will allow the team members to evaluate the students' learning.

In the second lesson, team members will explain the benefits of researching existing solutions to the problem. Students will explore similar products in a trade study, culminating in them producing a Pugh chart. Team members will explain the engineering concepts of risk, budgeting, and scheduling. They will engage the students by providing real-world examples such as the Boeing 737 Max 8. They will elaborate on these concepts, providing specific examples of integrating them into the design process. Students will explore these concepts further by creating a risk matrix, a budget, and a simple schedule. Team members will evaluate these and the students' Pugh charts.

In the third lesson, team members will explain the concepts of Key Decision Points, validation, and verification. They will elaborate on each of these, engaging the students by connecting them to their experience developing the RABeT System for the Micro-g NExT Challenge. Students will explore each of these concepts in relation to their own project by creating a ConOps, writing testing procedures, and defining success criteria. Team members will evaluate the students' work, providing feedback and guiding them in the process.

[10] Full lesson plan found in Appendix 4.9

In the fourth lesson, students will learn how to present their projects. Team members will explain the Preliminary Design Review (PDR) process of a proposal, elaborating on how this step affects the design process. Team members will also teach students about the steps after the PDR process.

In the fifth lesson, students will create a five-minute presentation describing their design, exploring the PDR process in relation to their project. Team members will act as the review panel for the students' proposals, evaluating them and exploring ways for them to improve. The student with the highest scoring proposal will receive a prize. Additionally, if current pandemic restrictions are lifted in such a way to allow for a hands on activity, the team will explore options to make the proposed tool a reality with the school faculty.

III. Administrative Section

3.1 Test Week Preference

Test Week I: June 7 -12, 2021

3.2 Mentor Request

The team is not currently collaborating on the project with a technical point of contact at NASA.

3.3 Institutional Letter of Endorsement

UNIVERSITY OF CALIFORNIA, BERKELEY

BERKELEY • DAVIS • IRVINE • LOS ANGELES • MERCED • RIVERSIDE • SAN DIEGO • SAN FRANCISCO SANTA BARBARA • SANTA CRUZ

PROFESSOR KARL VAN BIBBER
SHANKAR SASTRY CHAIR FOR LEADERSHIP & INNOVATION
DEPARTMENT OF NUCLEAR ENGINEERING
UNIVERSITY OF CALIFORNIA, BERKELEY
4107 ETCHEVERRY HALL
BERKELEY, CA 94720-1730

EXECUTIVE ASSOCIATE DEAN
ASSOCIATE DEAN OF RESEARCH
COLLEGE OF ENGINEERING
TELEPHONE: (510) 642 3477
email: karl.van.bibber@berkeley.edu

October 26, 2020

SUBJECT: LETTER OF ENDORSEMENT FOR NASA MICRO-g NExT CHALLENGE 2 – RABeT SYSTEM, UC BERKELEY

On behalf of the College of Engineering at the University of California, Berkeley, I am writing regarding the *Rapid Attachment Belt/Tool (RABeT) System* proposed by the Mechanical Engineering Competition (MEC) Team at Berkeley for NASA's Micro-g NExT Challenge 2, titled *Lunar Surface EVA Operations – Space Suit Attachment Quick Release System*. The University is aware of the team's interest in participating in this activity and endorses the team's involvement.

Sincerely,

Karl van Bibber

Karl van Bibber
Executive Associate Dean & Associate Dean for Research
College of Engineering, UC Berkeley

3.4 Statement of Supervising Faculty

UNIVERSITY OF CALIFORNIA
BERKELEY

Department of Mechanical Engineering

6159 Etcheverry Hall
Berkeley, CA 94720-1740
510 642-4901
510 642-5539 *fax*
hkt@berkeley.edu
http://design-nano.berkeley.edu

October 28, 2020

To whom it may concern:

As the faculty advisor for an experiment entitled **"Rapid Attachment Belt/Tool (RABeT) System"** proposed by a team of undergraduate students from the University of California, Berkeley, I concur with the concepts and methods by which this project will be conducted. I will ensure that all reports and deadlines are completed by the student team members in a timely manner. I understand that any default by this team concerning any Program requirements (including submission of final report materials) could adversely affect selection opportunities of future teams from the University of California, Berkeley.

Sincerely,

Hayden Taylor
Assistant Professor of Mechanical Engineering

3.5 Statement of Rights of Use

As a team member for a proposal entitled "Rapid Attachment Belt-Tool (RABeT) System" proposed by a team of undergraduate students from the University of California, Berkeley, I will and hereby do grant the U.S. Government a royalty-free, nonexclusive and irrevocable license to use, reproduce, distribute (including distribution by transmission) to the public, perform publicly, prepare derivative works, and display publicly, any data contained in this proposal in whole or in part and in any manner for Federal purposes and to have or permit others to do so for Federal purposes only.

As a team member for a proposal entitled "Rapid Attachment Belt-Tool (RABeT) System" proposed by a team of undergraduate students from the University of California, Berkeley, I will and hereby do grant the U.S. Government a nonexclusive, nontransferable, irrevocable, paid-up license to practice or have practiced for or on behalf of the United States an invention described or made part of this proposal throughout the world.

Trey Riddle - President, Project Lead

Pranit Mohnot - Vice President

Anthony Moody - Secretary, Design Lead

Wendy Yang - Outreach Chair

Todd Russell - Proposal Editor in Chief

Kevin Cheng - Member

Dr. Hayden Taylor - Faculty Advisor

3.6 Funding and Budget Statement

Items	Costs
Materials and Supplies	
3D Filament	$50.00
Aluminum	$70.00
Stainless Spring Steel	$20.00
Manufacturing Costs	
Machine Shop	$400.00
Travel	
Flights ($550 * 6)	$3,300.00
Hotel ($200/(room*night) * 3 rooms (double occupancy) * 6 nights)	$3,600.00
Ground Transportation ($650/week + $150 gas and parking)	$800.00
Food ($40/day * 6 students * 7 days)	$1,680.00
Other Expenses	
Shipping	$50.00
Team Shirts ($25 * 12)	$300.00
Total	$10,270.00

Potential Sources of Funding:

Student Senate Discretionary Funds - as needed
ASUC Student Opportunity Fund (Educational Opportunities) - $2000 per student for travel
Corporate Sponsorship - Donated time and materials for manufacturing

Financial Representative:

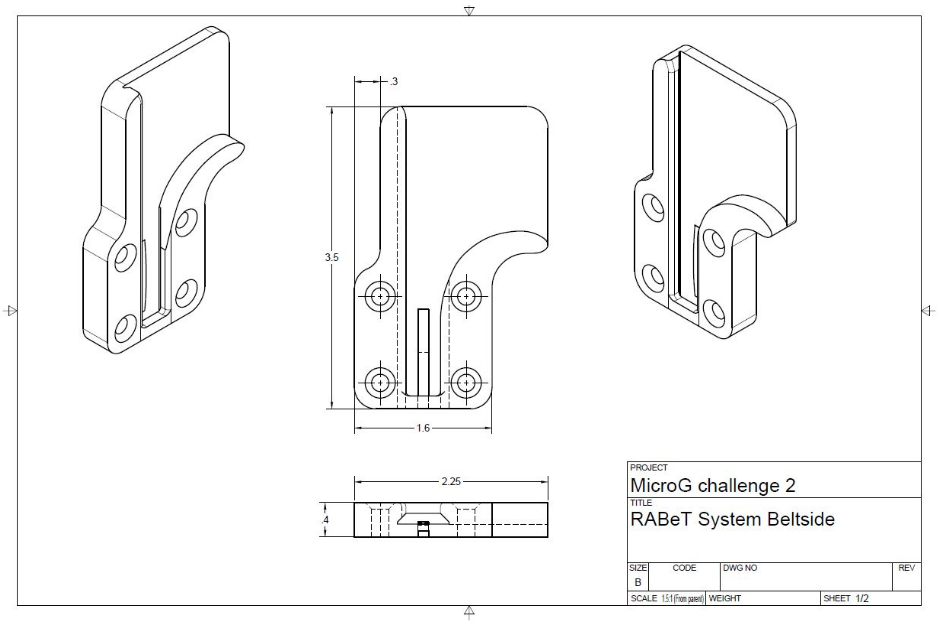
Deepak Sharma
Coordinator, Student Organization Advising and Leadership Development
deepak_sharma@berkeley.edu

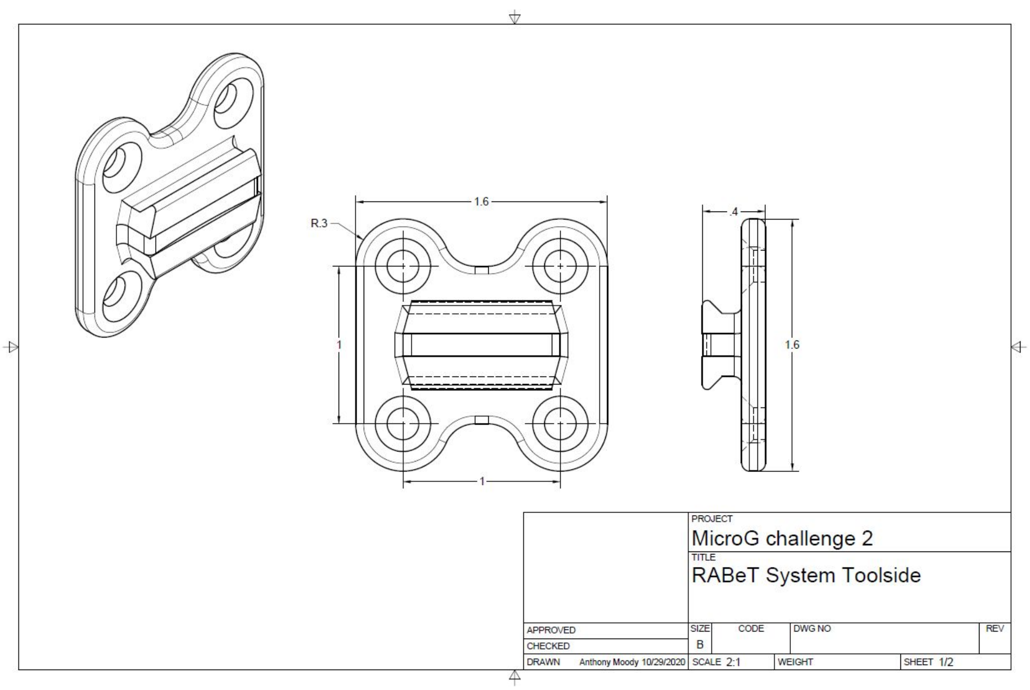
3.7 Parental Consent Forms

No team members are under eighteen years of age.

IV. Appendix

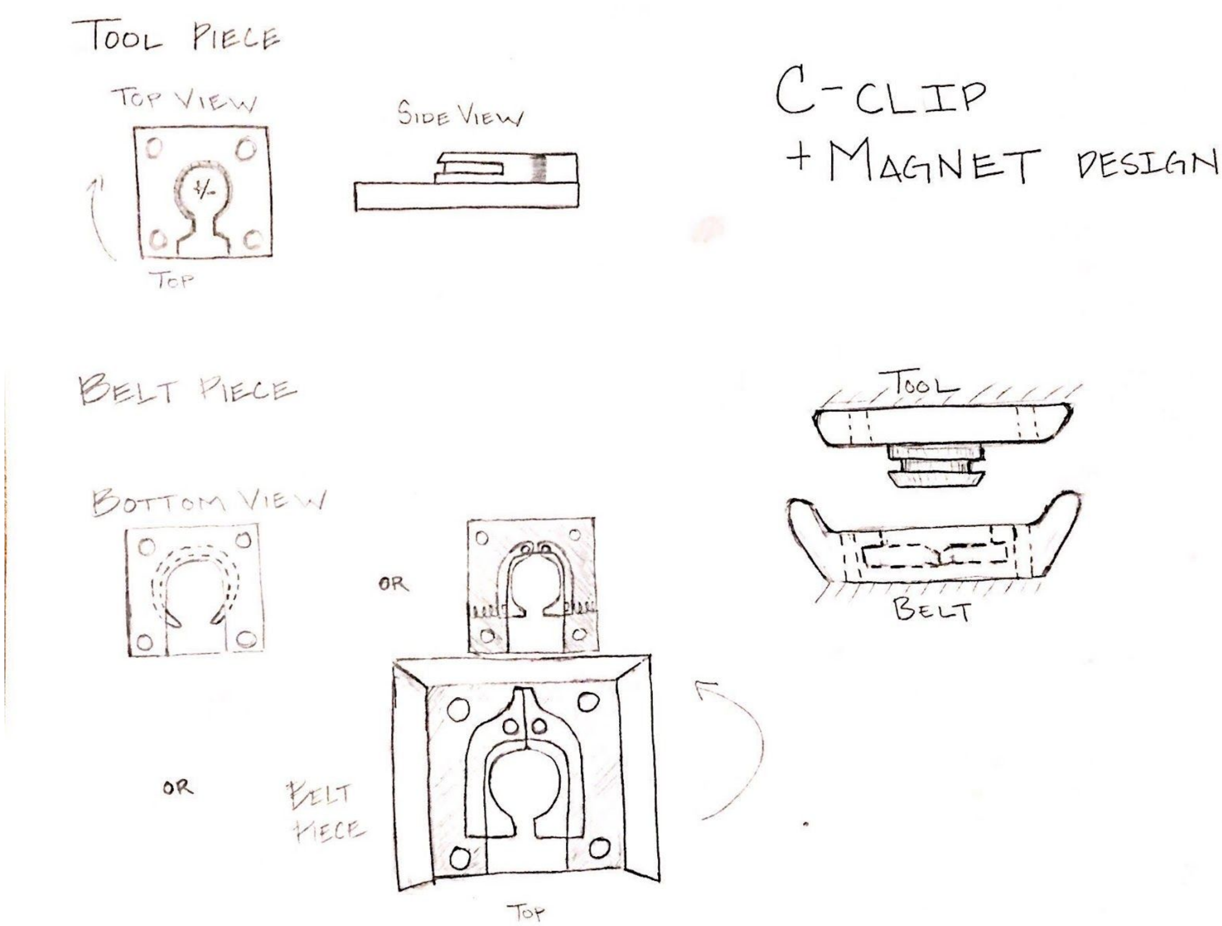
4.1 Additional Technical Drawings



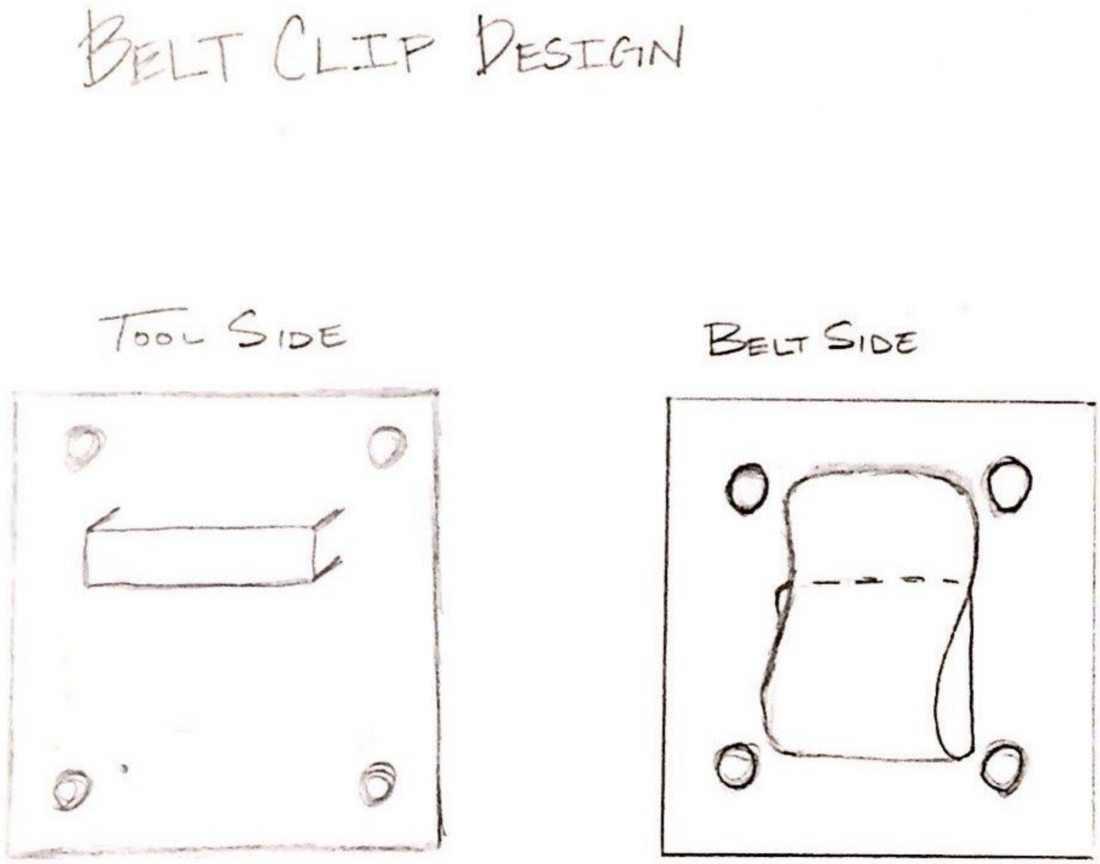


4.2 Initial Concepts Sketches

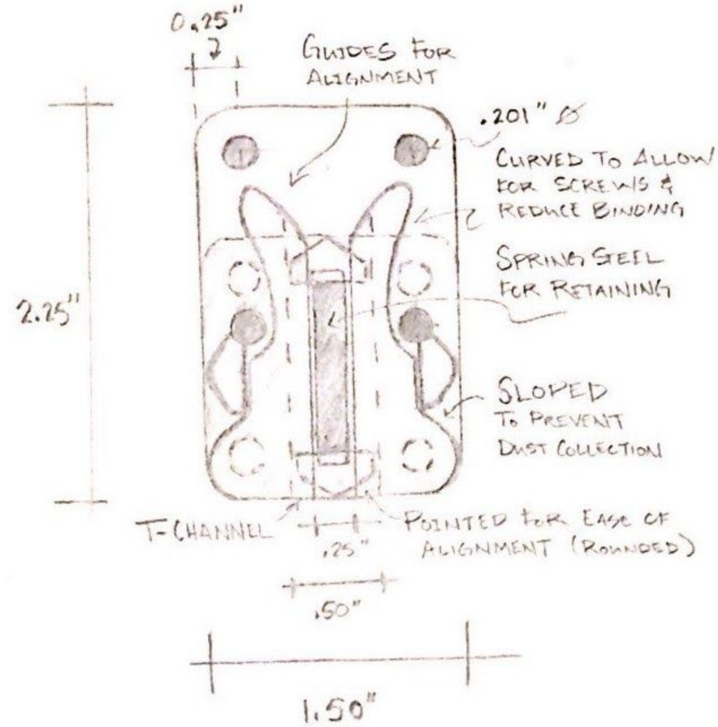
4.2.1 C-clip plus Magnet Design



4.2.2 Belt Clip Design

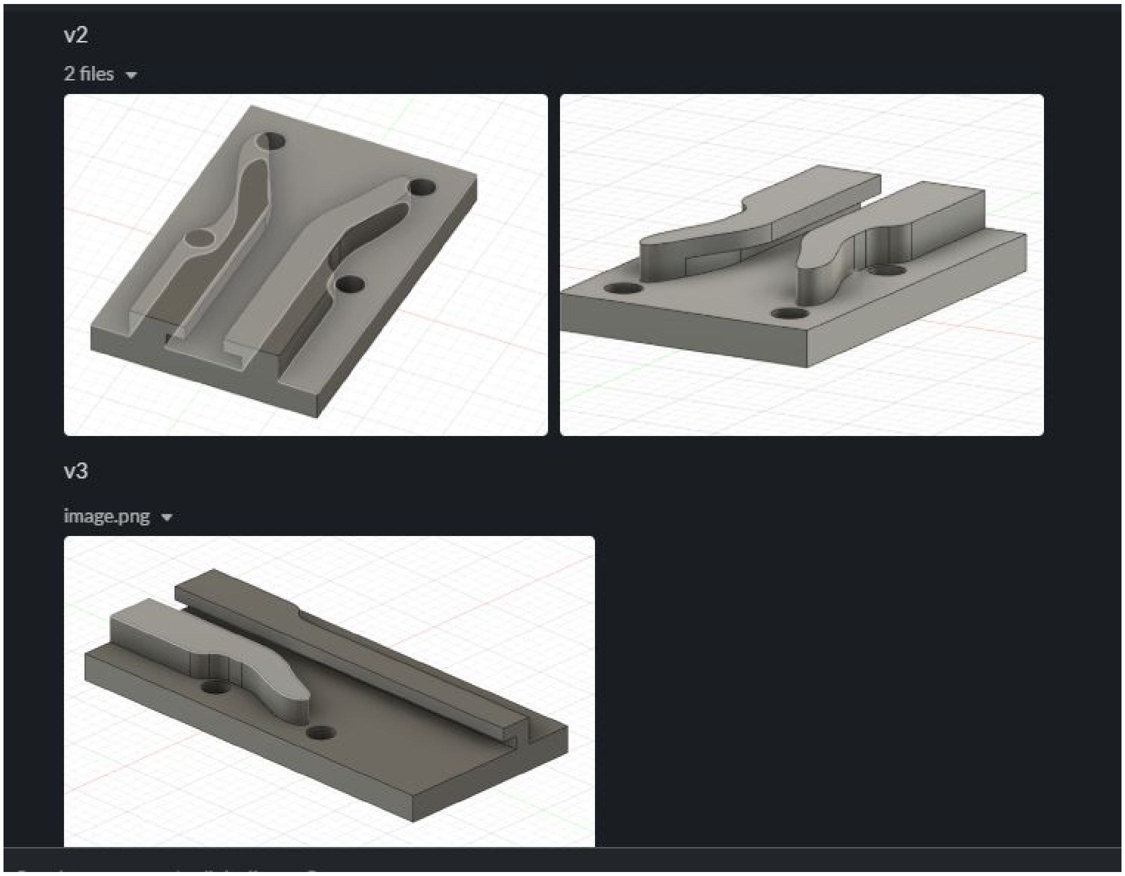


4.2.3 RABeT 1

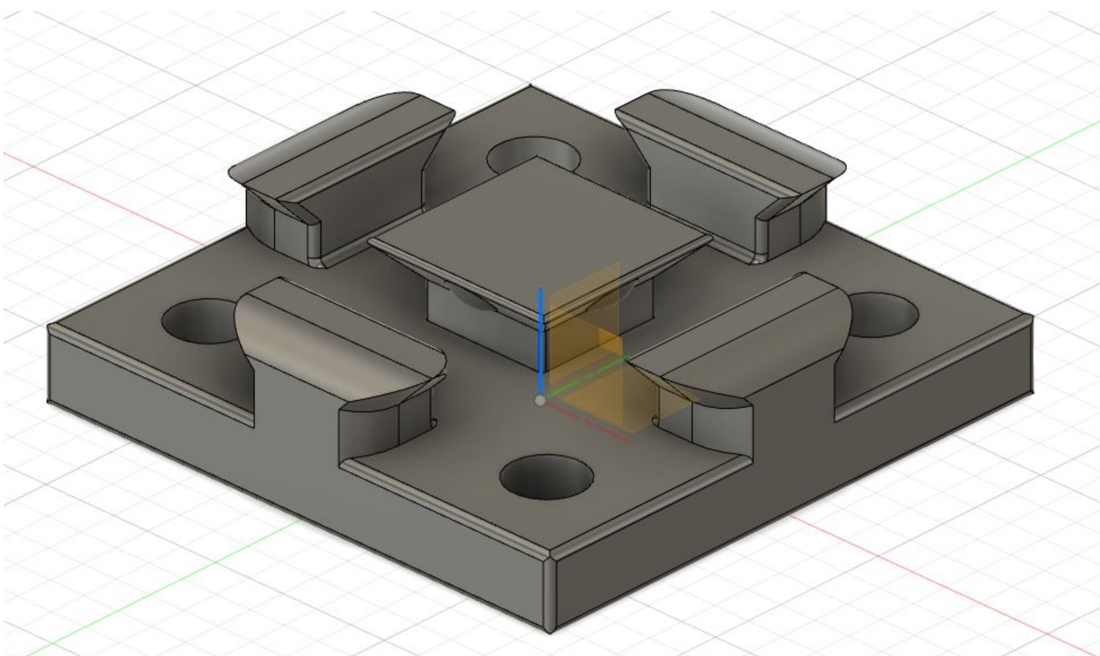


RAPID ATTACHMENT BELT/TOOL SYSTEM
(RABeT)

4.2.4 RABeT 2 & 3 Beltside



4.2.5 RABeT 2 Toolside

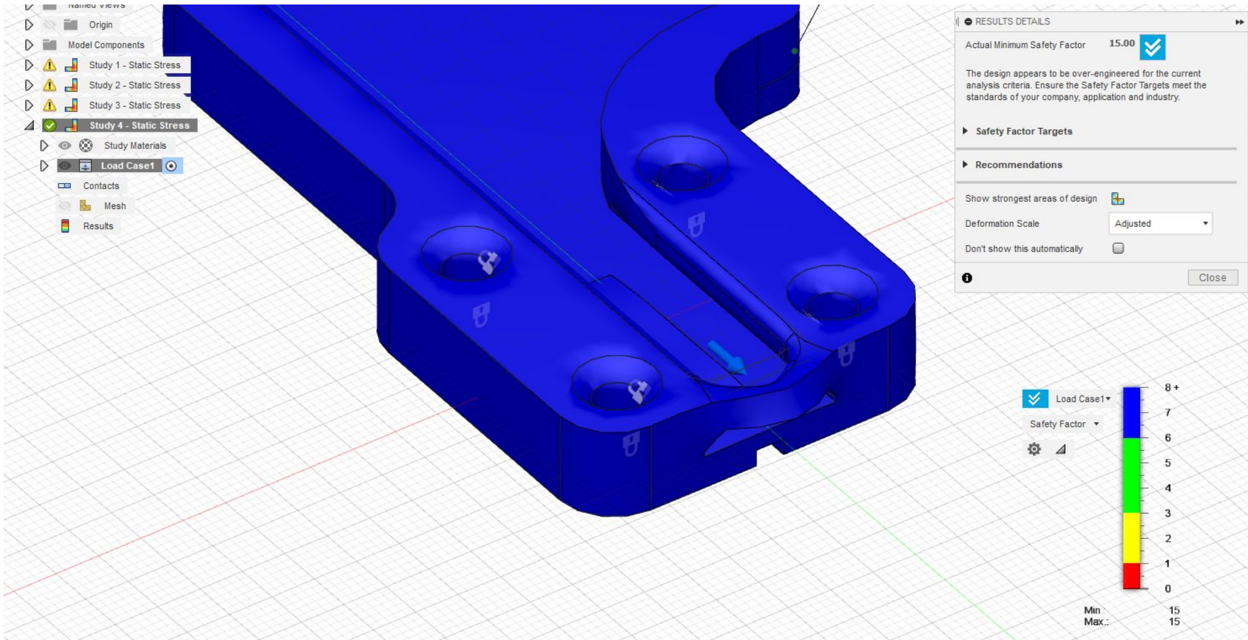


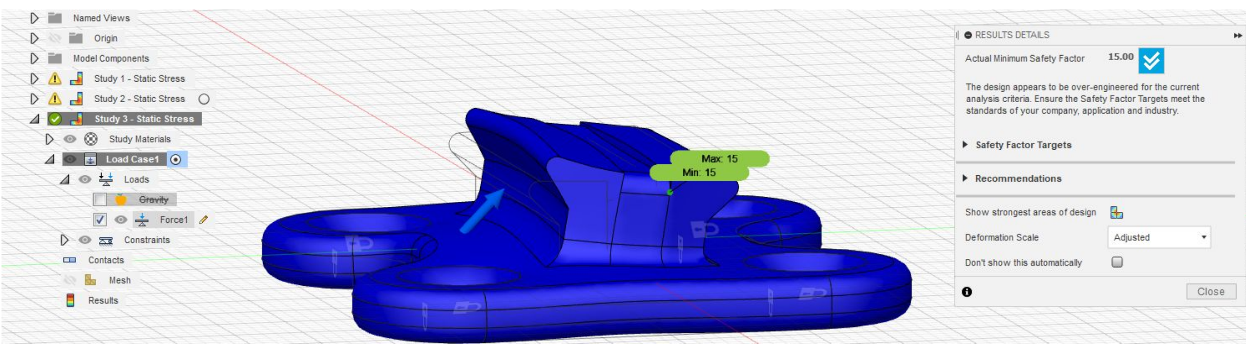
4.3 Trade Study Pugh Chart

	Weight	Belt Clip (#1)	C-Clip/Magnet (#2)	RABeT (#3)
Ease of Actuation	5	5	5	5
Safety	5	4	4	4
Tool Stability	5	2	4	5
Tool Security	5	4	5	4
Dust Tolerance	5	5	1	4
Strength	4	2	3	5
Size	4	4	5	4
Design Complexity	3	5	3	4
Weight	3	5	4	4
Manufacturability	2	3	2	3
Part Cost	1	4	2	3
Total	-	164	154	179

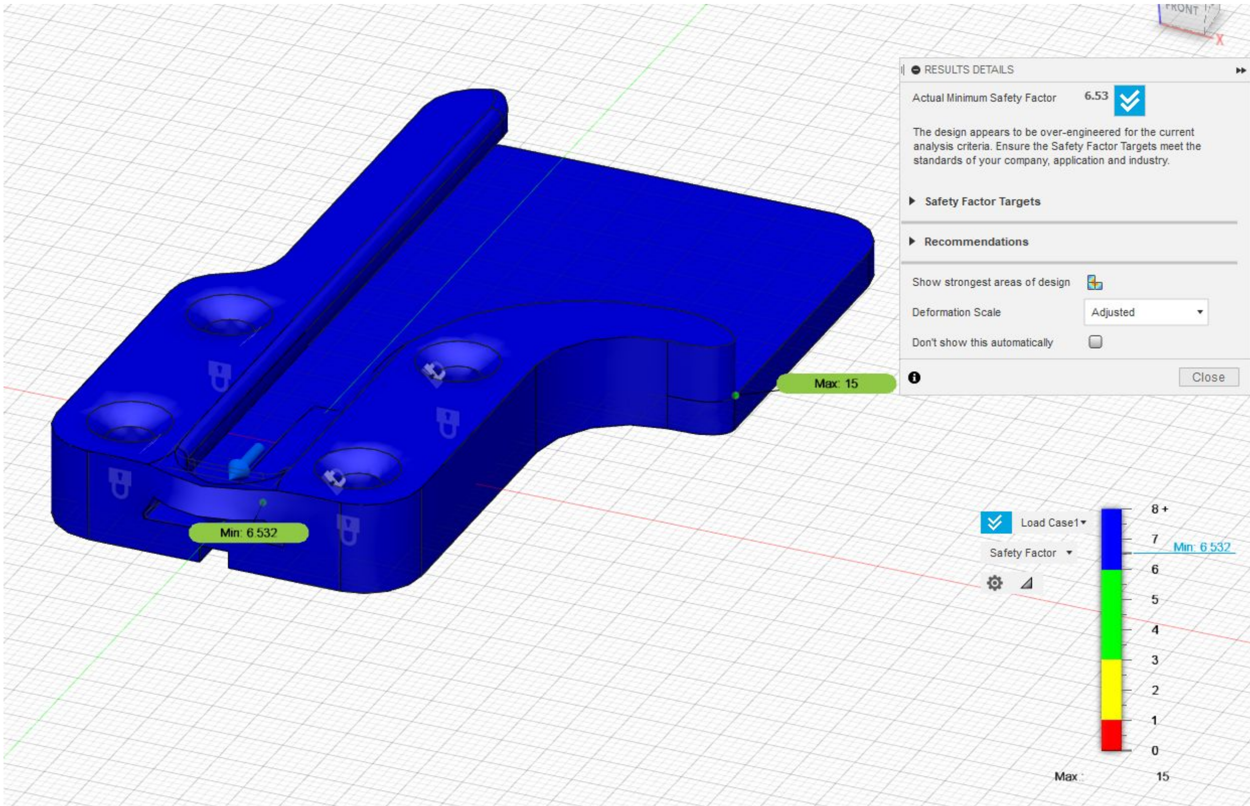
4.4 Finite Element Analysis (FEA)

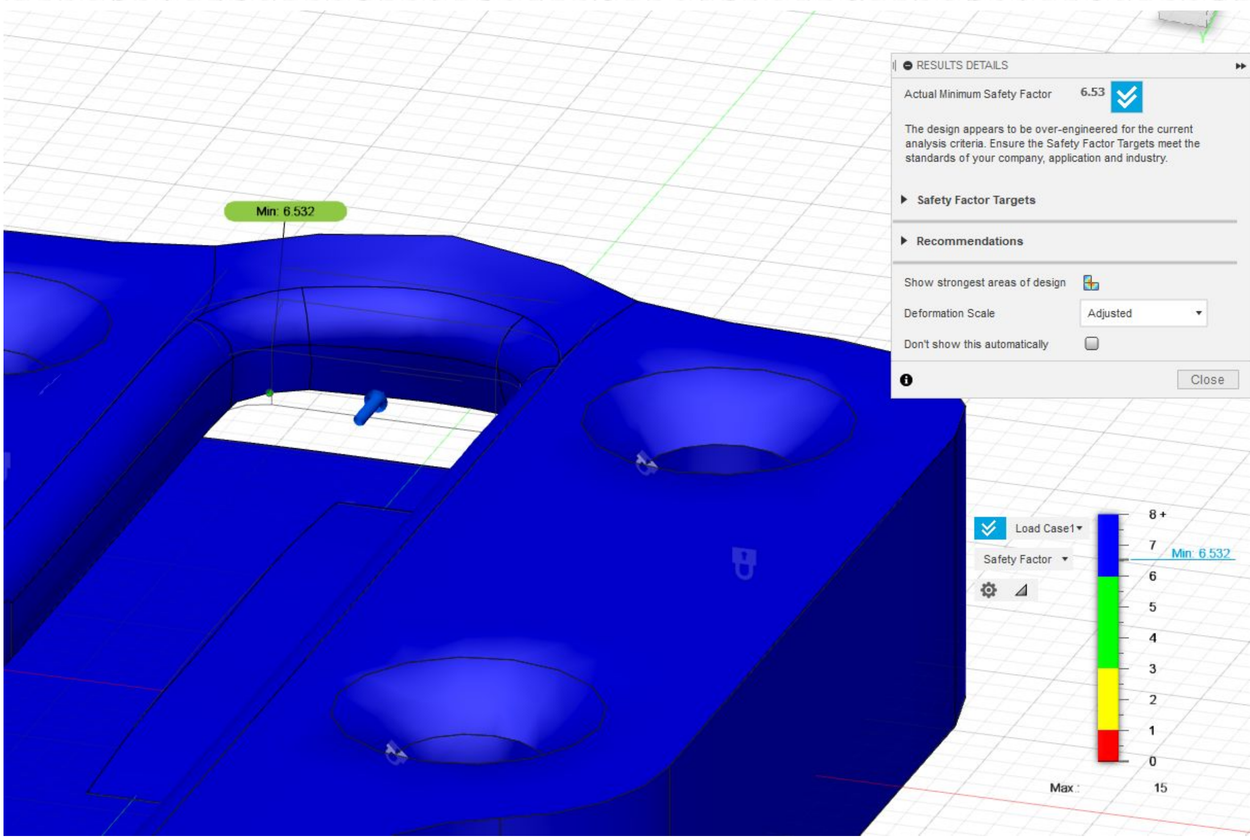
4.4.1 15 lb Load (Earth's gravity)





4.4.2 45 lb Load (Earth's gravity)





4.5 FMEA Risk Matrix

	25%+	A5	B5	C5	D5	E5
	10-25%	A4	B4	C4	D4	E4
	5-10%	A3	B3	C3	D3	E3
	1-5%	A2	B2	C2	D2	E2
Likelihood	0-1%	A1	B1	C1	D1	E1
Rubric		Minimal Impact to Mission	Small Effect on Mission Success	Moderate Effect on Mission Success	Significant Effect on Mission Success	Critical Mission Failure
		Mission Impact				
	Description of mission impact:	Redundant part failure. No impact on scientific mission. Minor damange to tool. CONOPS is not violated	Minimum impact for violation of CONOPS. Mission may operate in off-nominal conditions but largely fulfills CONOPS (e.g. tool isn't attaching correctly).	Buckle unable to fullfill key parts of mission without damage to tool	Buckle unable to fullfill key parts of mission, resulting in damage to the tool and other vehecles	Buckle is unable to perform duty. Lost of buckle and/or tool. Life-threatening damage to astronauts.

4.6 Press Release

Todd Russell
MEC Team at Berkeley
1 (916) 833-7559
Russellt@berkeley.edu

UC Berkeley Students Selected for Phase 2 in NASA Challenge

Berkeley, CA: As NASA shoots for the moon with their Artemis program, they continue to keep their roots on Earth by enlisting the help of university students around the nation to develop tools to help the astronauts on their missions. Composed of five separate design challenges, NASA's Micro-G Neutral Buoyancy Experiment Design Team (Micro-g NExT) competition challenges students to design, test, and propose solutions to these challenges.

A group of UC Berkeley students from the Mechanical Engineering Competition (MEC) Team at Berkeley are competing in NASA's Micro-G NExT challenge, and they were one of [number to be inserted once known] teams selected to participate in Phase Two.

The MEC Team is participating in the Space Suit Attachment Quick Release System challenge to design a quick release system which easily attaches tools to the astronaut's utility belt.

When notified of their selection for the next round, MEC Team president Trey Riddle said, "I'm so proud of the team and all the hard work they put into the proposal. They had many sleepless nights and really pulled together to make this project a success. I can't wait to go to JSC with them to test our system in the NBL."

Composed of six team members, the MEC Team chose NASA's Micro-G NExT challenge as their inaugural competition as a team. Hayden Taylor, an Assistant Professor of Mechanical Engineering and the faculty advisor for the MEC Team, said [quote to be inserted] when asked about the quick success of the team.

When approaching the challenge, the MEC Team looked towards common objects such as a bicycle helmet clasp for inspiration. Once they had a multitude of possible solutions to the problem, they explored each option further, comparing the pros and cons of each. Then, they consolidated their multiple solutions into one which took the best parts of each design.

They refined this singular design further, subjecting their model to stress testing to simulate the challenges it will face on the surface of the moon such as continuing to function even in the presence of lunar regolith.

As one of the teams selected for the next round, the MEC Team will travel to NASA's Neutral Buoyancy Laboratory (NBL) at the Johnson Space Center in Houston for professional divers to test their tool in a simulated microgravity environment.

For more information on the MEC Team's journey, you can like their page on Facebook.

4.7 Letters of Interest

4.7.1 St. Ignatius Parish School

St. Ignatius Parish School
3245 ARDEN WAY - SACRAMENTO, CA 95825 - 916-488-3907

DATE: October 26, 2020

TO: Mechanical Engineering Competition Team
At U.C. Berkeley

FROM: St. Ignatius Parish School

SUBJECT: Class Outreach and Presentation

Thank you for agreeing to present what you have learned while competing in the Micro-G NExt Challenge. St. Ignatius Parish School is delighted to have you lead a workshop teaching science and engineering principals to our students.

Students at St. Ignatius Parish School thrive in a dynamic and vibrant educational environment, where curiosity and creativity drive rigorous academic pursuits. Your workshop will be an ideal addition to our science curriculum.

As we discussed, you will be presenting to our Upper Science classes, consisting of 6th through 8th graders.

We are excited for you to share how you went about designing, building, and testing a tool and how it will address space exploration challenges, as well as guiding our students through a similar design process in your lunar design workshop.

We look forward to chatting with you and finalizing the details.

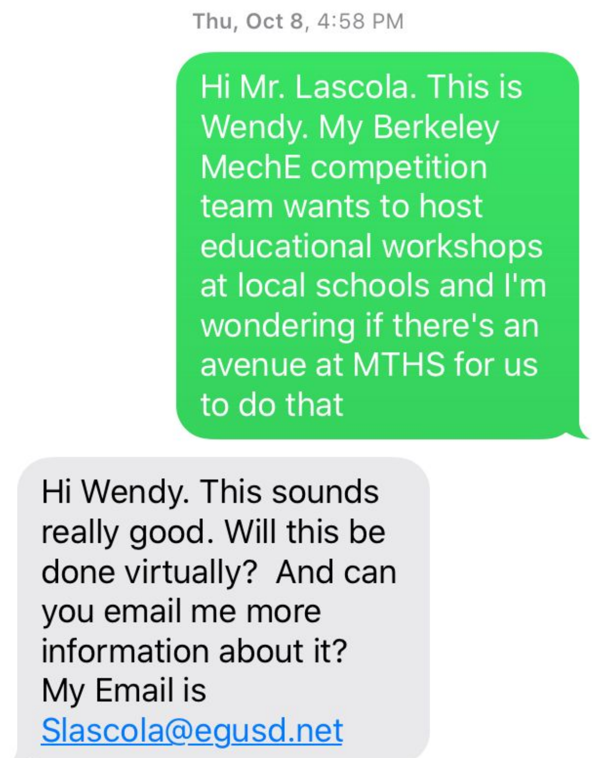
Sincerely,

Patricia Kochis
Principal

Serena Carleton
Upper Grade Science Teacher

4.7.2 Monterey Trail High School

Initial communication



Emails regarding lesson plan

Preliminary Lesson Plan

4 messages

Wendy Yang <xiaorui.yang@berkeley.edu> Tue, Oct 13, 2020 at 7:49 PM
To: slascola@egusd.net

Hi Mr. Lascola,

Linked here is the lesson plan I'm drafting. Currently I am still writing the details and the presentation. Can you review the outline and tell me about what you think about it?

Thanks
Wendy

--
Wendy Yang
Class of 2021
Mechanical Engineering
University of California, Bekreley

Salvatore Lascola at Monterey Trail HS <SLascola@egusd.net> Wed, Oct 14, 2020 at 6:31 AM
To: Wendy Yang <xiaorui.yang@berkeley.edu>

Hi Wendy.
I'll review and get back to you.

Thank you

Sal Lascola
Monterey Trail HS
Industrial Technology Dept.
Design and Technology Academy Coord.

From: Wendy Yang <xiaorui.yang@berkeley.edu>
Sent: Tuesday, October 13, 2020 7:49 PM
To: Salvatore Lascola at Monterey Trail HS <SLascola@egusd.net>
Subject: Preliminary Lesson Plan

CAUTION – EXTERNAL SENDER

[Quoted text hidden]

Salvatore Lascola at Monterey Trail HS <SLascola@egusd.net> Wed, Oct 21, 2020 at 7:54 AM
To: Wendy Yang <xiaorui.yang@berkeley.edu>

Hi Wendy.
Thank you for connecting with us to be part of your college work. After reading your proposal, I have

some questions and a few comments.

1. Is this to be a one-day workshop/presentation to high school students or do you envision delivering/teaching this to a group of students over multiple days?

2. If this is a multi-day workshop, how many students are you proposing take part in these lessons? And is it designed to be done during our school hours or after school?

3. Do you have any incentives for the students?

4. What are the desired outcomes your MechE group at Berkeley are trying to attain by delivering this workshop?

As you know we are currently working/teaching within a distance learning platform, and I have had to make serious modifications to my curriculum to account for this new learning environment. We are now only teaching 55-minute periods 4 days per week and a 30 min period on Mondays. We no longer have 90+ minute class periods, so there has been a reduction in the content we deliver. This means that I will be unable to implement any new material, like this workshop, within the class period.

If this is to be administered in an after-school workshop, I think our students will need an incentive to take part in it. They too are very stressed and any additional homework they have to complete will most likely not be looked upon as favorable.

Although this sounds like I may not want to take part in this workshop, I just want to express our current situation and the difficulties with implementing any new curriculum to our students.

Thank you

Sal Lascola
Monterey Trail HS
Industrial Technology Dept.
Design and Technology Academy Coord.

From: Wendy Yang <xiaorui.yang@berkeley.edu>
Sent: Tuesday, October 13, 2020 7:49 PM
To: Salvatore Lascola at Monterey Trail HS <SLascola@egusd.net>
Subject: Preliminary Lesson Plan

CAUTION – EXTERNAL SENDER

Hi Mr. Lascola,
[Quoted text hidden]

Wendy Yang <xiaorui.yang@berkeley.edu> Fri, Oct 23, 2020 at 4:19 PM
To: Salvatore Lascola at Monterey Trail HS <SLascola@egusd.net>

Hi Mr. Lascola,

Thank you for reading our lesson plan outline. Our plan with this workshop is that it is delivered over 5 days to small groups (~20) of high school students interested in engineering. The content of the workshop is self contained in the workshop and requires no other content. The workshop is a simplified version of a NASA undergrad workshop that requires an application to attend. Our plan is that students will sign up if they want to attend, but I must concede that we have not considered how to engage students outside of people who want to pursue engineering. The terms of NASA competition states that we're allowed to host the workshop outside of school semesters.

Our goal with the workshop is to teach basics of system engineering/project management, a skill that is important in college level engineering courses, internships, research projects, and engineering careers, but it is usually not taught in class. I've noticed that people who entered engineering at Berkeley usually already have that background from doing engineering competitions like FIRST in middle/high school, and I know not all schools offer that kind of opportunity. This competition in particular requires a heavy knowledge of system engineering in the proposal process, and I have the job of teaching this process to members of my team. NASA requires the outreach part to be related to the competition, so I thought we should do a lesson plan based on the process itself.

Thank you for your time
Wendy
[Quoted text hidden]

Communication confirming interest

Good morning Wendy. Yep. I got your email.

Do you think you and your group can create promotional information that would excite the kids to sign up? We also need to determine the days and times the workshops will be happening. I'll send our school schedule so we can figure that out.

4.8 Email Template

Dear [name]:

The Mechanical Engineering Competition Team at Berkeley would like to invite you and your students to participate in the team's outreach program as part of the NASA Micro-g NExT proposal challenge. The Micro-g NExT proposal challenge (https://microgravityuniversity.jsc.nasa.gov/about-micro-g-next.cfm) is an annual challenge for college undergraduates to participate in design challenges related to the US space program in order to further national interest and gain engineering experiences.

[Plan for middle school]
Our goal is to reach out to students and inspire them to pursue a career in STEM. Our lesson plan is a short engineering workshop related to lunar development. Students will first learn about the science and challenges involved in designing for a lunar environment, then design a simulated lunar enclosure using materials found at home that protects the inhabitants inside from environmental harm. Students then justify how their design protects the inhabitants and why their design is better than other designs. The entire workshop can be done virtually and can be wholly conducted by members of the team.

[Plan for high school]
Our goal is to reach out to students and inspire them to pursue a career in STEM. Our lesson plan is a 5 days project management workshop that teaches students how to effectively manage a project. Students will learn about the fundamentals of systems engineering, then apply what they have learned to design an apparatus that can operate in an underwater (reduced gravity) environment against abrasion to collect trash. The lesson includes a documentation process that culminates in a student presentation of their designs. The student(s) with the highest presentation score will be able to receive a prize from the team. The workshop is conducted virtually and can be conducted wholly by the team. The point of the lesson is to provide students with engineering knowledge that can prepare them for engineering careers.

The details of the lesson plan are attached below. We hope you accept our invitation. Please don't hesitate to contact us with any questions. Thank you for your time and consideration.

Sincerely,
[Name]
Mechanical Engineering Competition Team at Berkeley

4.9 Lesson Plans

4.9.1 Middle School (6-8)

Lunar Design Lesson Plan for 6-8 Graders

(Berkeley MEC Team)

(Developed by Todd Russell and Wendy Yang)

Abstract

The purpose of this lesson plan is to teach middle school students the science and engineering behind designing for a lunar mission. Students will learn about the science of the environment of the moon, what engineering challenges face human lunar missions, and the current missions planned for the moon. Then students will perform a hands-on design activity to apply what they have learned. The workshop will take an hour to conduct.

1 Development

This lesson plan is developed as part of the NASA Micro-g NExT Proposal Challenge. The plan of this lesson is to describe challenges of designing for a lunar mission that the team encounters while completing the challenge, formulate it in a way digestible for middle school students, and culminating in a hands-on activity performed by students. The goal of the lesson is to inspire students to consider space industry as a career path.

2 Lesson Information

2.1 Big idea

Accounting for different aspects of lunar environment is essential to human exploration of the moon

2.2 Objectives

- Students will learn about the moon
- Students will learn how to design a tool to withstand a simulated lunar environment

2.3 Skills Developed

- Problem Solving
- Brainstorming
- Research

2.4 California Educational Standards Addressed

- **CTE.EA.B.4.0:** Understand the concepts of physics that are fundamental to engineering technology.
- **MS-ESS1-2:** Earth's Place in the Universe.

- **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

2.5 Materials

- Sand
- Cardboard
- Glue
- Tape

3 Lesson Outline

3.1 The challenges of the moon

In this section, the students learn about the history of human lunar exploration and the challenges the moon posses for any future human exploration.

3.1.1 The moon

Students should learn about key conditions of the moon that makes it different than living on Earth. This includes

- Lack of air
- Lunar dust
- Reduced gravity
- Radiation
- Cold

All of these makes human settlement on the moon difficult.

3.1.2 Past human exploration

Students should learn about past human explorations and their limitations (why don't we go to the moon anymore).

3.1.3 Future human exploration

Students should learn about what lunar exploration is planned for the future (like CLPS) and what opportunities they present.

3.2 The activity

In this section, students use what they have learned to construct an apparatus that shields a human model from the effect of lunar dust.
The students should consider what they now know about the moon and brainstorm what someone living on the moon need to consider. Educators should prompt them to ask

- What does a human need to *live*?
- What does the Earth provide that the moon doesn't have?
- How do we replicate that on the moon?

After this exercise, the students can begin to plan and construct their apparatus. Consider setting some constraints to their design, such as

- It must have a way for the people inside to get out
- It must be under a certain size
- It must be under a certain weight

After the student construct their apparatus, the student should test the effectiveness of their apparatus by pouring sand (carefully) over it. The effectiveness is measured by how much sand gets inside the apparatus.

3.2.1 Deliverable

- a model of the apparatus
- a short report detailing what the student did, how does the apparatus protect the humans inside

4 Further Development

The challenge section may include materials on the political challenges which face the US space program.

4.9.2 High School (9-12)

Water Design Lesson Plan for 9-12 Graders

(Berkeley MEC Team)
(Developed by Todd Russell and Wendy Yang)

Abstract

The purpose of this lesson is to teach high school students the process of an engineering project. Students will learn how to develop specifications, how to develop a tool to conform to specification, and how to evaluate and test the produced tool. The length of this workshop is one week. Students will be assigned their task after the first lesson, and then a checkoff will be performed at the beginning of the next lesson. Students will present their work at the end of the week in the form of a presentation.

1 Development

This lesson plan is developed as part of the NASA Micro-g NExT Proposal Challenge. The lesson is formulated as a simplified version of the proposal challenge with a comparable engineering question that incorporates the challenges of designing for a lunar environment but on an Earth environment. The goal of the lesson is to equip high school students with basic engineering management skills that are needed in a college and career-engineering environment. The team recognizes that not all high schools were able to provide this kind of education to students; this creates an uneven playing field in higher education. Thus, this lesson plan is created to fill this gap and inspire students to pursue engineering.

2 Lesson Information

2.1 Big idea

Effective engineering project management is essential to project success

2.2 Objectives

- Students will learn how to plan an engineering project
- Students will learn how to effectively manage an engineering project

2.3 Skills Developed

- Problem Solving
- Brainstorming
- Research
- Time Management

2.4 California Educational Standards Addressed

- **CTE.EA.B.7.0:** Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance.
- **CTE.EA.C.2.0:** Understand the effective use of engineering design equipment.
- **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

3 Lesson Outline

3.1 Lesson 1: Foundations

In this lesson, students will learn the importance of systems engineering and how to develop good guiding foundations for the engineering project.

3.1.1 Terms

- ConOps (Concept of Operations): a timeline that describes how the user of the tool will use the tool
- Stakeholder: individual or group that is affected by the product or the project
- Top Level Requirements: description of how a system or a part must operate.

3.1.2 Procedure

The lesson begins by giving students an overview of system engineering. The students should understand the key role of systems engineering on any projects from individual to team projects.
In the NASA Systems Engineering Handbook, systems engineering is defined as

"An interdisciplinary approach for the design, realization, technical management, operations, and retirement of a system."

The role of a systems engineer is different than most perceptions of engineers, but effective systems engineering is vital to project success. A systems engineer looks at the overall big picture of the project and sees how different aspects of the system interconnect in order to deliver a system which meets requirements while working within constraints.
A "system" is defined as a

"Combination of elements that function together to produce the capability required to meet a need."

Systems include small tools, houses, software, power plants, and the International Space Station. Systems engineering is utilized to solve many problems that are important today. In this lesson, an engineering problem pertaining to our environment is presented to the students:

Students shall develop a low-cost tool to be operated by divers cleaning trash out of the Sacramento River.

The engineering problem is purposely broad. Problems in the real world are not like homework or project assignments. It is up to the engineers to define what the systems should do and how to answer the engineering question. The engineer must keep in mind stakeholder expectations when they answer these key questions. Stakeholders include not just the user of the system, but also any other party whom the systems affect in any ways. Stakeholder expectations are sometimes not clearly defined in the engineering problem; engineers must research what they are. In other times, they are clearly defined, such as a space mission proposal solicitation, and they act as a constraint on the system. Students should establish stakeholder expectations for this problem:

- Who uses the tool?
- Who does the tool affects?
- Who benefits from the tool?
- How effective should the tool be?

Once the stakeholder expectations have been established, students should develop a concept of operation for the the tool. A ConOps helps to define how the system meets stakeholder expectations. A good ConOps outlines the usage of the system from beginning to systems disposal, and it also considers what happens to the system outside of its intended usage. Students should develop a clear concept of operation graphics. The rest of the project is based on the ConOps; thus, it is very important to ensure a good ConOps has been developed at this stage.
Once a good ConOps has been developed, top level requirements are determined based on ConOps. Good top level requirements have defined numbers and can be verified with testing or analysis. Requirements are "shall" statements for the operation of the system and should be non-negotiable without waivers. Educators should prompt students to ask

- What operation environment does the tool have to operate in?
- What problems does the operation environment present?
- What problem does the tool solve and how does it solve it?
- How will the tool be maintained and stored?

3.1.3 Homework

- Students shall develop a concept of operations for their tool
- Students shall establish stakeholder expectations based on the problem
- Students shall write top level requirements for their tool

3.2 Lesson 2: Process

In this lesson, students will learn how to research and evaluate different solutions, and how to create an effective budget and schedule for the project.

3.2.1 Terms

- Trade Study: a study comparing different methods
- Pugh Chart: a trade study chart
- Risk matrix: a matrix which evaluates risks the project has.
- Gantt Chart: a scheduling timeline containing tasks and personnel assignment.
- State of the Art: highest level of current development

3.2.2 Procedures

After the development of top level requirements, it is time to research how to meet those requirements. In most engineering problems, similar problems may already have active solutions, or there are already products that are solving the problem, but it just doesn't meet certain expectations. Engineers should research existing solutions to the problem so they do not reinvent the wheel. This is so the final product will be—at the very least—an improvement on the existing solutions. This is also where the engineer will research manufacturing processes, potential shortcomings, and risks.
The research result is gathered to perform a trade study. A trade study compares different solutions to select the best solution which is viable with the given constraints. Students need to ask how they want to improve upon the existing state of the art products:

- Does it address a problem the current products do not?
- Does it cost less?
- It it more durable?

A trade study is usually compiled into a Pugh chart. Different aspects of the existing products are compiled into a weighted chart to pick out the best solutions. It is used for manufacturing methods, materials, costs, etc.—any aspect which is relevant to the tool and the previously written top-level requirements. The chart helps engineers begin drafting their design, and it is used to justify why the engineers chose the design they did.
It is also important to consider any risks the tool may have. These includes injury, damage, improper handling, etc. It should encompass everything from the best case scenario to the worst case scenario. The students should ask:

- If the tool was handled perfectly based on the instruction, what kind of injuries could still occur?
- If the tool was handled improperly, what kind of injuries would occur?
- What kind of risks can occur just based on chance?
- What kind of risks do environmental factor and repeated use cause?
- How can the tool fail?

The risks are compiled into an failure mode and effects analysis (FMEA). An FMEA lists all the risk the tool can cause and ranks them based on likelihood and risk of injury. The ranks are based on a risk matrix, which assigns a alphanumerical code based on a combination of probability and damage. Besides ranking the list, an FMEA also includes a way to mitigate the risks via design and other methods such as training and safety protection.

Another important aspect of engineering is budgeting. Budgeting is a real world constraint on engineering alongside physics and politics; otherwise, engineers would already be running amok on the moon. At this stage, a broad budget estimate based on trade studies and the FMEA should be created with refinements added later on. Each line of the budget should have a justification on why this line costs this much and why it is necessary. A proper budget estimate can have a range, but should not have broad ranges without sufficient justification.

Lastly, engineers need to consider scheduling. A proper schedule gives time for the engineers to train, design, manufacture, test, meet key milestones, and have enough "slack" to give leeway to the process in case anything unforeseeable happens (such as a pandemic, for example). In a team environment, a schedule also includes personnel assignments and additional assignments in case the assignment personnel drops off the project due to unforeseen issues. A good schedule includes good deadlines, proper time allocated to each of the engineering process, milestones, deliverables, and proper assignments. In the case of a team, proper reasonings are usually given for each personnel assignment. Schedules are usually compiled into a Gantt chart.

3.2.3 Homework

- Students shall research existing solutions to the problem
- Students shall create a Pugh chart for manufacturing methods and existing solutions
- Students shall create a risk matrix
- Students shall draft a budget
- Students shall draft a simple schedule

3.3 Lesson 3: Design, Testing, Success

In this lesson, students will learn effective design process and how to develop tests for the tool they develop. They will then evaluate the success of their tool.

3.3.1 Terms

- Key Decision Point (KDP): when the decision holder decides if the project can progress to the next phase
- Validation: shows the system accomplishes its intended purpose
- Verification: shows the system complies with requirements

3.3.2 Procedure

Once the previous steps are finished, it is time to review them. This is a Key Decision Point in the engineering process which determines if the project can proceed to the design phase. If the project passes the review, the engineers can proceed to the design phase.

This is a "preliminary" design phase, meaning the design is defined enough to establish a baseline, but it is not finalized. A finalized design would be created after a review of the preliminary design (but will not

be covered in this lesson plan due to format constraints). Usually, a preliminary design (and all the steps before it) is enough to meet a solicitation call. If the proposal wins the solicitation/gets the grant, then the engineers would move to further stages.
The design should meet top level requirements and ConOps. Justifications have to be made for how it meets the requirements. In larger projects, it is possible to request a waiver on certain requirements for certain parts of a system; the decision makers of the projects would give the approval. This process is how the design is "verified" to comply with the requirements and actually address the needs. A way to verify the design is to create a compliance matrix. Top level requirements are listed and checked off based on the design. It is good to give numbers in the justification:

- What safety factor does the design have?
- How much stuff does the design scoop?

In the process of design, new risks and new needs may arise. The FMEA, budget, and schedule should be updated to reflect these new additions. Generally, those elements serve as the baseline for future evaluation in future phases, so it is essential to make the necessary revisions.
Besides making the preliminary design, engineers also consider how to validate the tool itself. The tool is useless if it doesn't meet the requirement in actuality. This means designing tests on the tool. A good test validates the tool in a simulated, controlled environment, with easily quantifiable data. Test should be based on previously written top level requirements. Consider asking the following questions:

- What measuring tool should be used?
- What data can be collected?
- What does a tool failure mean?

The test results are what qualifies the performance of the tool to meet "success criteria". Success criteria describes the outcome of usage of the system that would meet the requirements of the stakeholders. It is what engineers use to sell the tool to stakeholders. It is defined in measurable terms that can be compared against the test result.

3.3.3 Homework

- Students shall draft a simple design
- Students shall write testing procedures
- Students shall define success criteria

3.4 Lesson 4: Presentation and Actualization

In this lesson, students will learn how to present their project and the next steps of the project cycle.

3.4.1 Terms

- Quad chart: A one page chart that describes the proposal
- PDR: Preliminary Design Review

3.4.2 Procedure

In major engineering projects and solicitations, the previous segment is combined into a Preliminary Design Review (PDR) of a proposal. The proposal is evaluated on its strength, usually by a review board, before the project is given the approval to proceed to future steps. If the proposal does not pass the PDR—either because it fails some requirements or there are better proposals—the project does not proceed at all. In some projects, this means the proposal must be revised and reviewed again. In other projects and solicitations, the proposal dies. In engineering careers and grad school, this process is essential and common. A well written proposal can ensure grant funding for years to come and create opportunities for essential research. Thus, common proposals may range from a few pages up to hundreds of pages for major projects.
Due to constraints, this exercise will not ask students to compile the previous section into a well written proposal. Instead, we ask the students to create a short, five minute presentation describing their design. The presentation should include the question the design is addressing, requirements, scheduling with milestones and deliverables, the design itself, a budget, tests, and success criteria. The presentation will be graded, but *only* what is in the presentation within the time limit can be graded. Anything else will be disregarded.
If the proposal passes the PDR, the next step is to finalize and manufacture the design. Previous baseline plans are revised and updated. The schedule should be adhered to. Then, once the tool is fabricated, the project moves to the next step. Testing is performed on the tool, operation plans are written, and safety checks are performed. When those are finished, the tool can move to its operation phase, which is being used by its stakeholders to perform its duty. And finally, at the end of the tool's operation, a decommission plan is formulated.

3.4.3 Homework

- Students shall organize their material into a five minute presentation

3.5 Lesson 5: Student Presentation

In this lesson, students will present their project. The best scoring presentation will receive a prize.

3.6 Procedure

In this final section, the role of the educator is to act as the review panel for the students' proposals. A rubric should be created to evaluate the students' proposals on a point scale system in order to give students feedback and grade the proposal. Upon completion, the students receive their feedback. The highest scoring proposal will receive a prize which acts as the grant of the project.

4 Further Development

The development of this lesson plan is constrained by the format of virtual lessons. It is possible to incorporate production elements into it. This will necessitate lengthening the lesson plan to include practical applications of other stages of systems engineering.

5 Further Reading

The lesson plan is based on the NASA Systems Engineering Handbook (https://www.nasa.gov/connect/ebooks/nasa-systems-engineering-handbook).

6 Addendum

Special thanks to ASU/NASA L'Space Academy for inspiring this lesson.